# STRUCTURAL SYSTEMS 1 *ARC 2713*

**STEEL BUILDING: COLUMNS & FOUNDATION**

Lecture 11

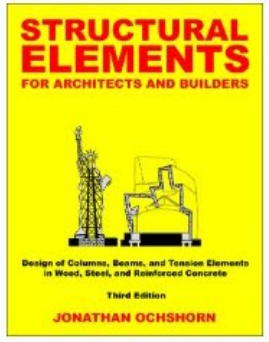
**1. COLUMN SELECTION for sizing and load to foundation**
**3. FOUNDATION parameters & configurations**
**4. STEEL BUILDING Reference File for Configuration Options & Parameters**

# STEEL BUILDING project

## PROJECT CALCULATORS: Steel Beams and Columns

OCHSHORN Steel Beam Calculator: https://jonochshorn.com/scholarship/calculators-st/example8.6/index.html

OCHSHORN Steel Column Calculator: https://jonochshorn.com/scholarship/calculators-st/steel-column-design/index.html




**Laterally-Supported Steel Floor Beam Calculator**

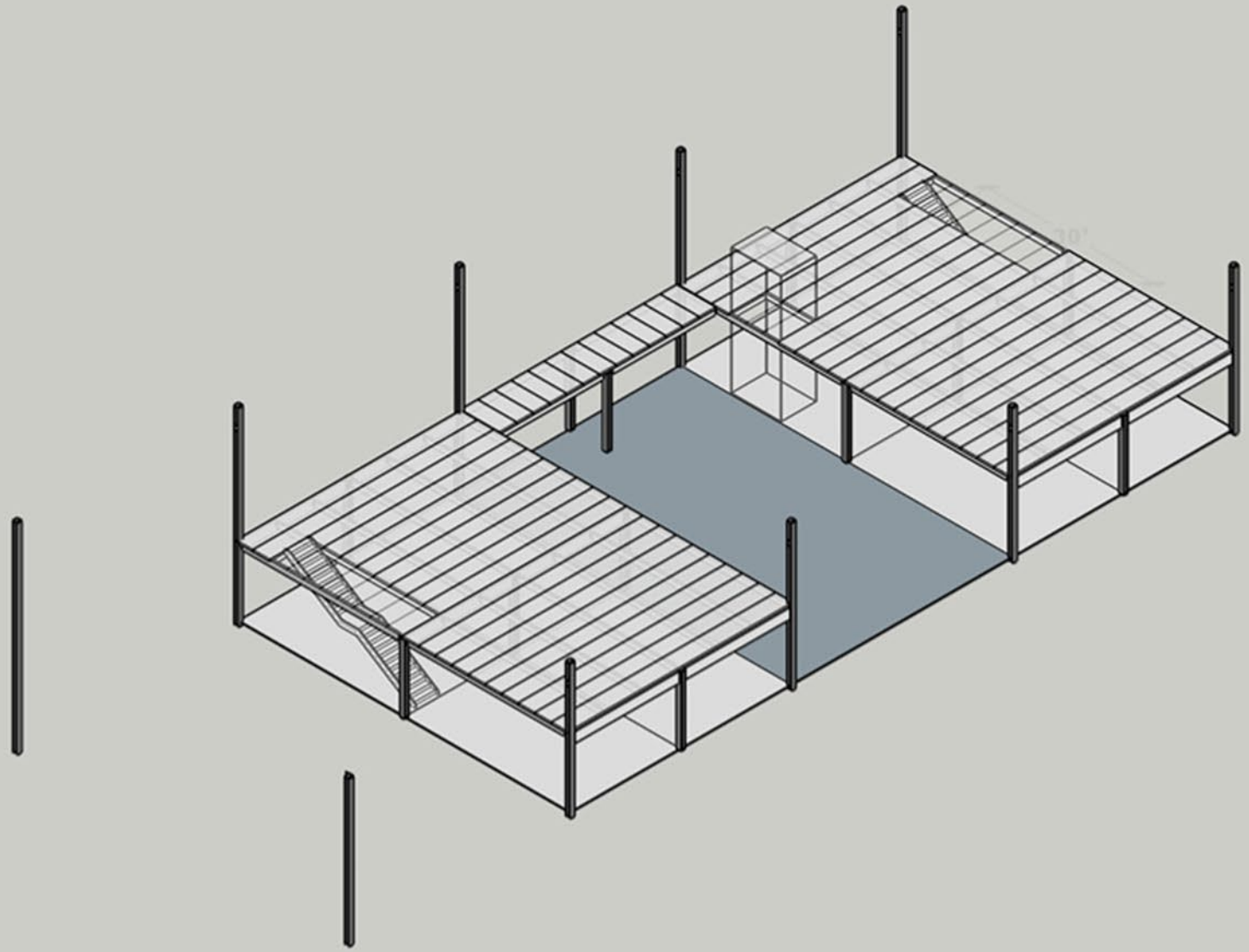
Jonathan Ochshorn

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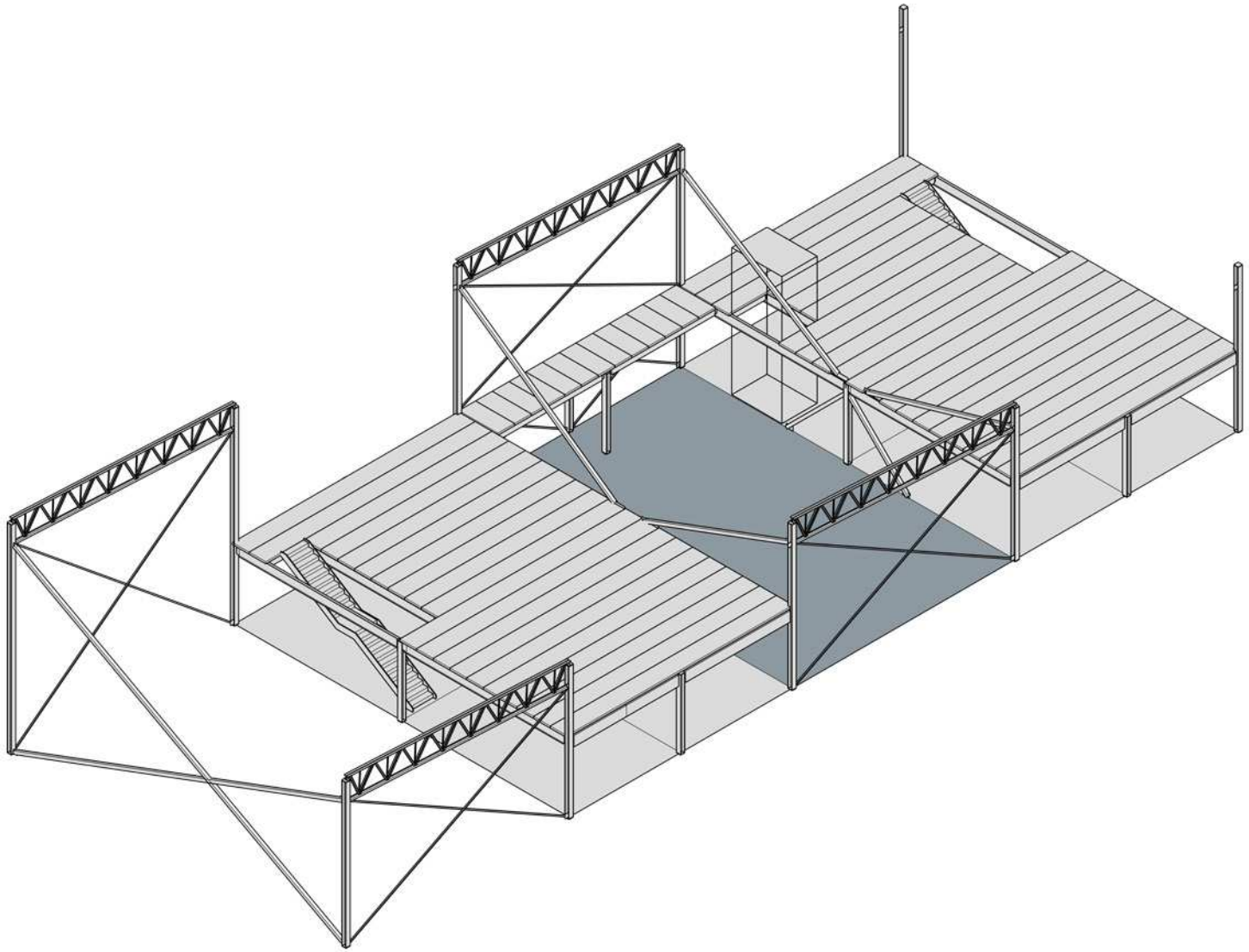
# STEEL BUILDING as LIBRARY

**PROJECT PARAMETER SELECTIONS:**

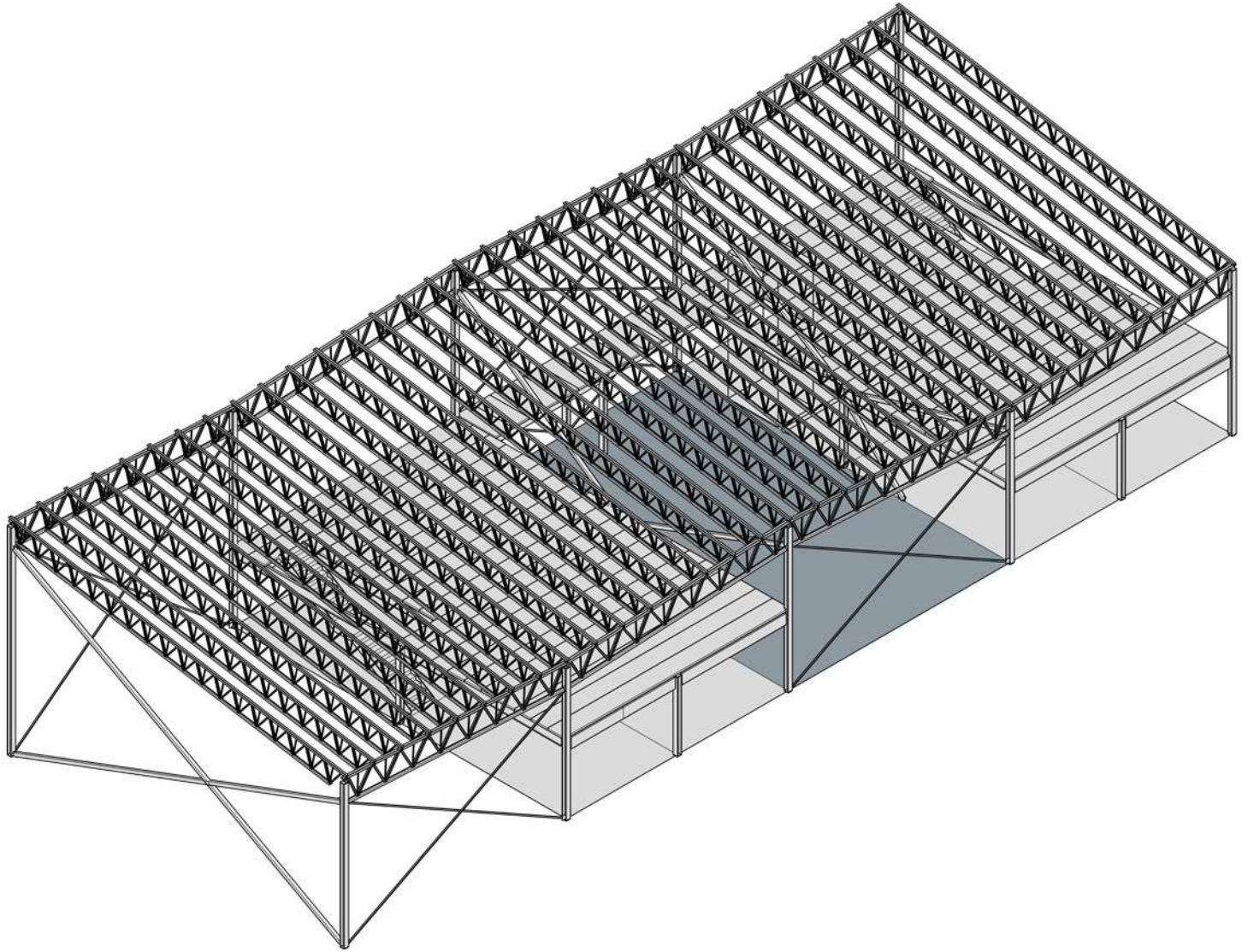
**FLOOR: MODEL 4060 R**
**ROOF: MODEL 4060 C**



# FLOOR STRUCTURE: MODEL 4060R



# LATERAL STRUCTURE: MODEL 4060R



**ROOF STRUCTURE: MODEL 4060C**

## FLOOR DECK

**3 ½” Slab Depth (total depth)**
**20-gauge steel Composite Deck**
**with normal-weight concrete**
(Composite steel deck self-weight: 2 psf)
(Concrete weight: 31 psf)

## LIBRARY FLOOR LOAD

150 psf: Live Load
15 psf: Dead Load:
2 psf: Deck self-weight
31 psf: Concrete weight
**198 psf: Floor Load on Floor Deck**

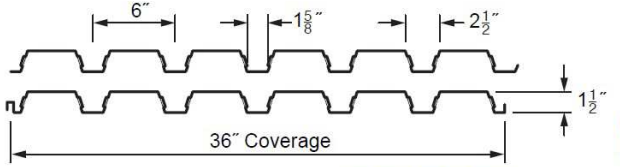
**Composite Deck spans 7.5’ maximum** (198 psf < 215 psf)

# 1.5CD, 1.5CDI Composite Deck

NORMAL-WEIGHT CONCRETE

PROPERTIES

| Gage | Thickness (in) | Cover (in) | Weight (psf) |
|---|---|---|---|
| 22 | 0.0295 | 3 | 1.63 |
| 20 | 0.0358 | | 1.98 |
| 18 | 0.0474 | | 2.62 |
| 16 | 0.0598 | | 3.30 |




ICC-ES APPROVED
REFER TO ESR-3818

ASD SECTION PROPERTIES

DESIGN STRENGTHS (No Concrete Fill)

| Gage | $F_y$ (ksi) | $A_s$ (in²/ft) | $I_D$ 1 Span (in⁴/ft) | $I_D$ 2+ Span (in⁴/ft) | $I_p$ (in⁴/ft) | | $S_p$ (in³/ft) | $S_n$ (in³/ft) | Mn,p/Ω (in-lb/ft) | Mn,n/Ω (in-lb/ft) | Vn/Ω (lb/ft) | *Rbe/Ω (lb/ft) | *Rbi/Ω (lb/ft) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 22 | 40 | 0.480 | 0.163 | 0.175 | 0.157 | | 0.176 | 0.183 | 4215 | 4391 | 2009 | 654 | 1181 |
| 20 | 40 | 0.582 | 0.205 | 0.213 | 0.201 | | 0.224 | 0.235 | 5375 | 5624 | 2428 | 932 | 1696 |
| 18 | 40 | 0.770 | 0.280 | 0.281 | 0.279 | | 0.304 | 0.315 | 7291 | 7553 | 3192 | 1557 | 2858 |
| 16 | 40 | 0.971 | 0.355 | 0.355 | 0.355 | | 0.390 | 0.395 | 9332 | 9454 | 3996 | 2381 | 4396 |

CONSTRUCTION CLEAR SPANS

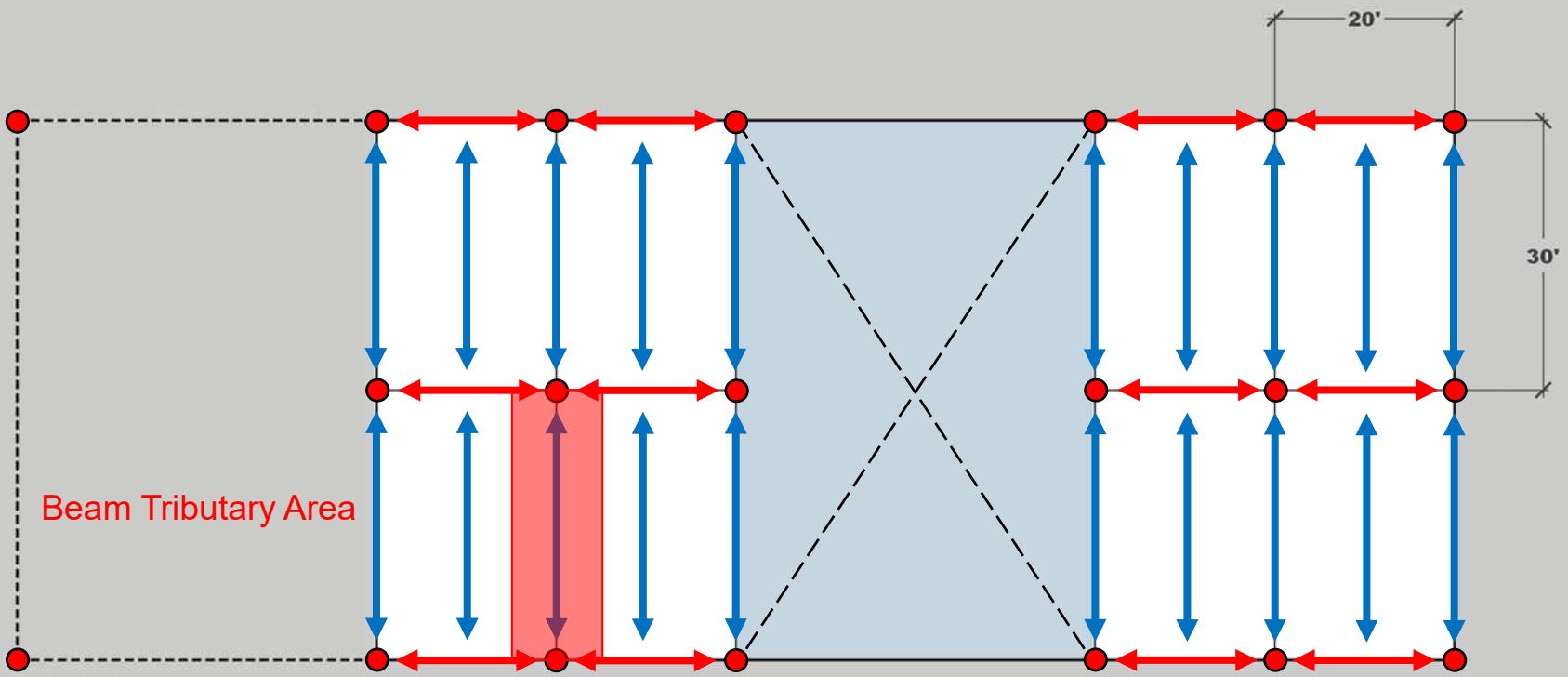
| Total Slab Depth (in) | Gage | Concrete Weight (psf) | Maximum Construction Clear Span (ft - in) | | | | Concrete Volume (ft³/ft²) |
|---|---|---|---|---|---|---|---|
| | | | Single | Double | Triple | Cantilever | |
| 3 1/2 | 22 | 31 | 5 - 9 | 6 - 9 | 6 - 10 | 2 - 0 | 0.213 |
| | 20 | | 6 - 9 | 8 - 0 | 8 - 1 | 2 - 5 | |
| | 18 | | 8 - 4 | 9 - 8 | 9 - 11 | 3 - 1 | |
| | 16 | | 9 - 8 | 10 - 9 | 11 - 1 | 3 - 8 | |

SLAB DESIGN - ALLOWABLE SUPERIMPOSED UNIFORM LOADS AND MINIMUM SDI SLAB REINFORCEMENT

Normal-Weight Concrete (γc = 145 pcf, f'c = 3000 psi, n = 9)

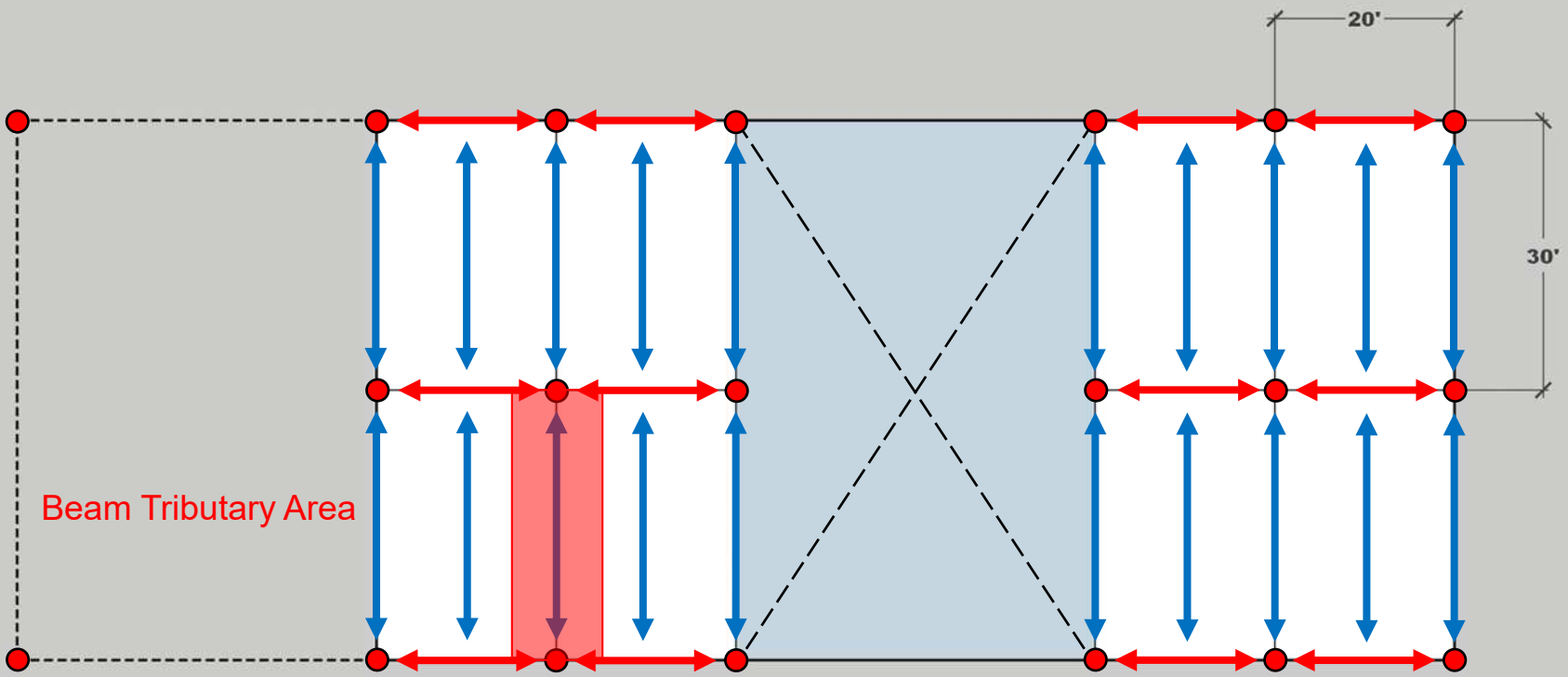
| Total Slab Depth (in) | Gage | Allowable Superimposed Uniform Load (psf) Clear Span (ft - in) | | | | | | | | | Recommended Welded Wire Fabric | Wire Area (in²/ft) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 6 - 0 | 6 - 6 | 7 - 0 | 7 - 6 | 8 - 0 | 8 - 6 | 9 - 0 | 10 - 0 | 11 - 0 | | |
| 3 1/2 | 22 | 294 | 246 | 207 | 176 | 151 | 130 | 113 | 85 | 65 | 6x6 - W1.4xW1.4 | 0.028 |
| | 20 | 354 | 297 | 251 | 215 | 185 | 160 | 139 | 106 | 82 | | |
| | 18 | 400 | 386 | 328 | 281 | 243 | 212 | 185 | 139 | 105 | | |
| | 16 | 400 | 400 | 400 | 349 | 301 | 251 | 212 | 154 | 116 | | |
| 4 | 22 | 370 | 310 | 262 | 223 | 191 | 165 | 143 | 109 | 83 | 6x6 - W1.4xW1.4 | 0.028 |
| | 20 | 400 | 375 | 318 | 272 | 234 | 203 | 177 | 136 | 105 | | |
| | 18 | 400 | 400 | 400 | 357 | 309 | 269 | 236 | 183 | 145 | | |
| | 16 | 400 | 400 | 400 | 400 | 385 | 336 | 295 | 229 | 172 | | |
| 4 1/2 | 22 | 400 | 377 | 319 | 272 | 234 | 202 | 175 | 133 | 103 | 6x6 - W1.4xW1.4 | 0.028 |
| | 20 | 400 | 400 | 387 | 331 | 286 | 248 | 216 | 167 | 130 | | |
| | 18 | 400 | 400 | 400 | 400 | 378 | 330 | 289 | 225 | 178 | | |
| | 16 | 400 | 400 | 400 | 400 | 400 | 400 | 363 | 285 | 228 | | |
| 5 | 22 | 400 | 400 | 378 | 322 | 277 | 240 | 208 | 159 | 123 | 6x6 - W2.0xW2.0 | 0.040 |
| | 20 | 400 | 400 | 400 | 393 | 339 | 295 | 258 | 199 | 156 | | |
| | 18 | 400 | 400 | 400 | 400 | 400 | 392 | 344 | 269 | 213 | | |
| | 16 | 400 | 400 | 400 | 400 | 400 | 400 | 400 | 341 | 273 | | |
| 5 1/2 | 22 | 400 | 400 | 400 | 374 | 322 | 279 | 242 | 186 | 144 | 6x6 - W2.0xW2.0 | 0.040 |
| | 20 | 400 | 400 | 400 | 400 | 394 | 343 | 300 | 232 | 182 | | |
| | 18 | 400 | 400 | 400 | 400 | 400 | 400 | 400 | 314 | 249 | | |
| | 16 | 400 | 400 | 400 | 400 | 400 | 400 | 400 | 398 | 319 | | |
| 6 | 22 | 400 | 400 | 400 | 400 | 367 | 318 | 277 | 212 | 165 | 4x4 - W1.4xW1.4 | 0.042 |
| | 20 | 400 | 400 | 400 | 400 | 400 | 392 | 342 | 265 | 208 | | |
| | 18 | 400 | 400 | 400 | 400 | 400 | 400 | 400 | 360 | 286 | | |
| | 16 | 400 | 400 | 400 | 400 | 400 | 400 | 400 | 400 | 366 | | |

**Notes:**
7. Welded wire fabric shown in the table above is the SDI minimum required for temperature and shrinkage (0.00075 x concrete area above top of deck).
8. Allowable Superimposed Uniform Loads shown in table above are based on the following criteria:
   - Unfactored service level loads, determined using SDI design method per ANSI/SDI C-2017 Standard for Composite Steel Floor Deck-Slabs.
   - Single span conditions without negative bending reinforcing over supports.
   - The presence of shear studs have not been considered, design strength may be increased if shear studs are used.
   - Slab deflection is limited to minimum of Clear Span/360 or 1" under service level superimposed loading.




**Beams spaced at 6.67’ o.c.**
**Beams span 30’ – 0”**
**Beam Load Calculation: (198 psf) ( 6.67’) = 1,320.66 plf**
**Beam Load = 1,321 plf**

**4060 R** **2nd Floor Plan option selected*…Beam Load is 1,321 plf***




**Beams spaced at 6.67’ o.c.**
**Beams span 30’ – 0”**
**Beam Load Calculation: (198 psf) ( 6.67’) = 1,320.66 plf**
**Beam Load = 1,321 plf**

# 4060 R 2nd Floor Plan option selected…*What is the size of the Beam?*

# PROMPT

Calculate the size of the lightest weight steel wide flange beam that spans 30 feet with a uniformly distributed load of 1,321 plf. The live load portion of the load is 1,000 plf. Limit Live load deflection to L/360.

# Find Beam size using Google Gemini

To determine the lightest standard ASTM A992 ($F_y = 50$ ksi) steel wide flange beam (W-shape), we evaluate both the **live load deflection limit** and the **flexural strength requirements** (assuming continuous lateral support to the compression flange).

## Step 1: Deflection Limit Criterion

**Span ($L$):** $30 \text{ ft} = 360 \text{ in}$

**Modulus of Elasticity ($E$):** $29,000 \text{ ksi}$

**Live Load ($w_L$):** $1,000 \text{ plf} = 1.000 \text{ k/ft} = \frac{1.0}{12} \text{ k/in}$

**Allowable Live Load Deflection ($\Delta_{L,\text{allow}}$):**

$$\Delta_{L,\text{allow}} = \frac{L}{360} = \frac{360 \text{ in}}{360} = 1.00 \text{ in}$$

Using the maximum deflection formula for a simple beam under uniform load:

$$\Delta_L = \frac{5 w_L L^4}{384 E I_x} \leq 1.00 \text{ in}$$

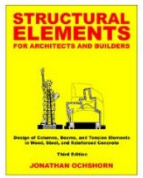
Solving for the required moment of inertia ($I_{x,\text{req}}$):

$$I_{x,\text{req}} = \frac{5 w_L L^4}{384 E \Delta_{L,\text{allow}}} = \frac{5 \left(\frac{1.0}{12} \text{ k/in}\right) (360 \text{ in})^4}{384 (29,000 \text{ ksi})(1.00 \text{ in})} = 628.45 \text{ in}^4$$

**jonochshorn.com**

Structural elements calculators

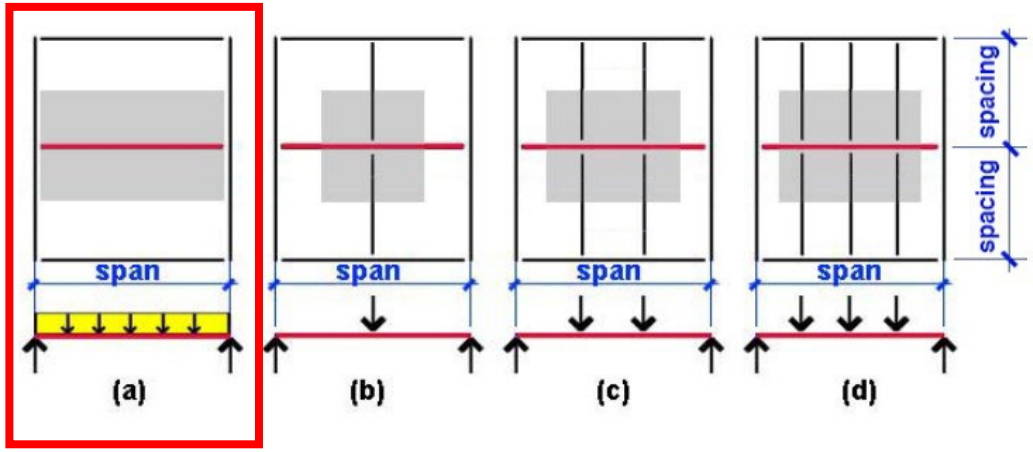
contact




# Laterally-Supported Steel Floor Beam Calculator

Jonathan Ochshorn

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## Uniformly-distributed Load

| span (ft) | spacing | |
|---|---|---|
| 30 | 6.67 | ft |

| ASTM designation and yield stress | live load (psf) | dead load (psf) |
|---|---|---|
| A992: Fy=50 ksi | 150 | 48 |

**(not used)**

60

***Yield stress of 50 ksi has been selected.***

**allow. live load deflection = span/**

**360**

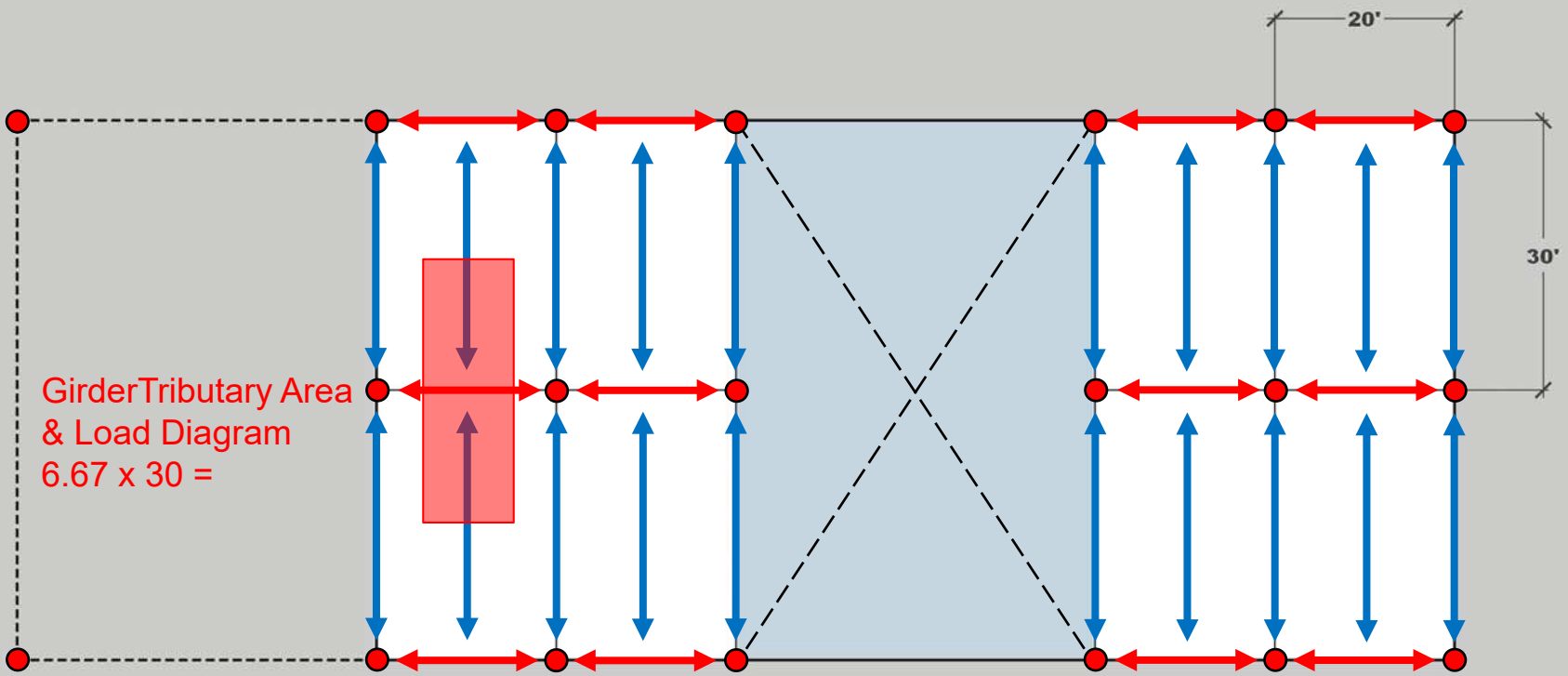
**allow. total load deflection = span/**

**240**

| Load type: | Mmax (ft-kips) | Vmax (kips) | Lightest choice |
|---|---|---|---|
| a) uniformly-distributed load | **149** | **19.8** | ***W21×44*** |
| b) load at mid-span | **149** | **9.9** | ***W18×35*** |
| c) loads at third points | **132** | **13.2** | ***W18×40*** |
| d) loads at quarter points | **149** | **14.9** | ***W18×40*** |

| Design criteria: | | reduction coeff: | live load used: |
|---|---|---|---|
| Use live load reduction? | no | **not used** | **150 psf** |
| Consider live load deflection? | yes | | |
| Consider total load deflection? | no | | |
| Consider shear stress? | yes | | |
| Consider bending stress? | yes | | |
| Lateral support: | continuous | | |

**Beam Size = W21 x 44**



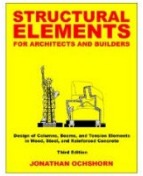

**Girders spaced at 30’ o.c.**
**Girders span 20’ – 0”**
**Girder Load Calculation: Live Load 150 psf + Dead Load 48 psf + DL beam 7 psf**
**Girder Load = 205 psf = LL 150 psf + DL 55 psf**

# 4060 R 2nd Floor Plan option selected…*What is the size of the Girder?*

**jonochshorn.com**

Structural elements calculators

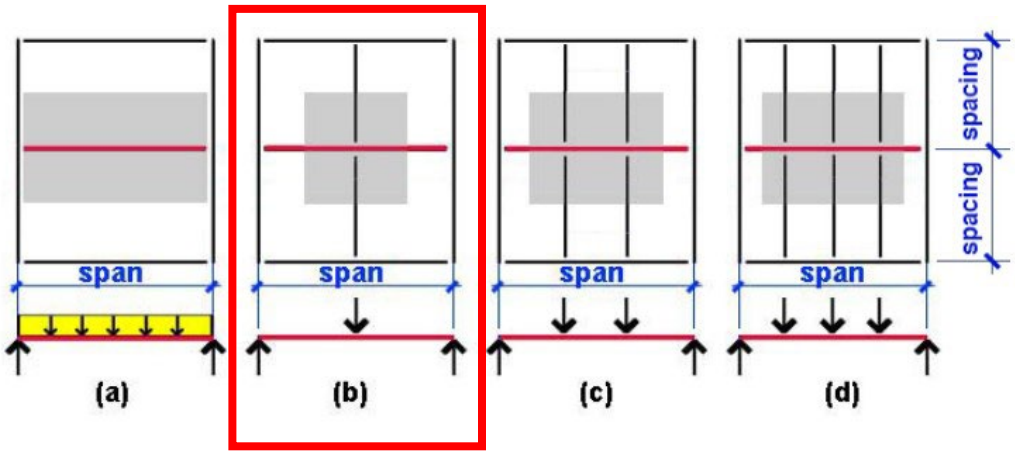
contact




# Laterally-Supported Steel Floor Beam Calculator

Jonathan Ochshorn

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**Girder Load at mid-span**

| span (ft) | spacing | |
|---|---|---|
| 20 | 30 | ft |

| ASTM designation and yield stress | live load (psf) | dead load (psf) |
|---|---|---|
| A992: Fy=50 ksi | 150 | 55 |

**(not used)**

60

***Yield stress of 50 ksi has been selected.***

**allow. live load deflection = span/**

**360**

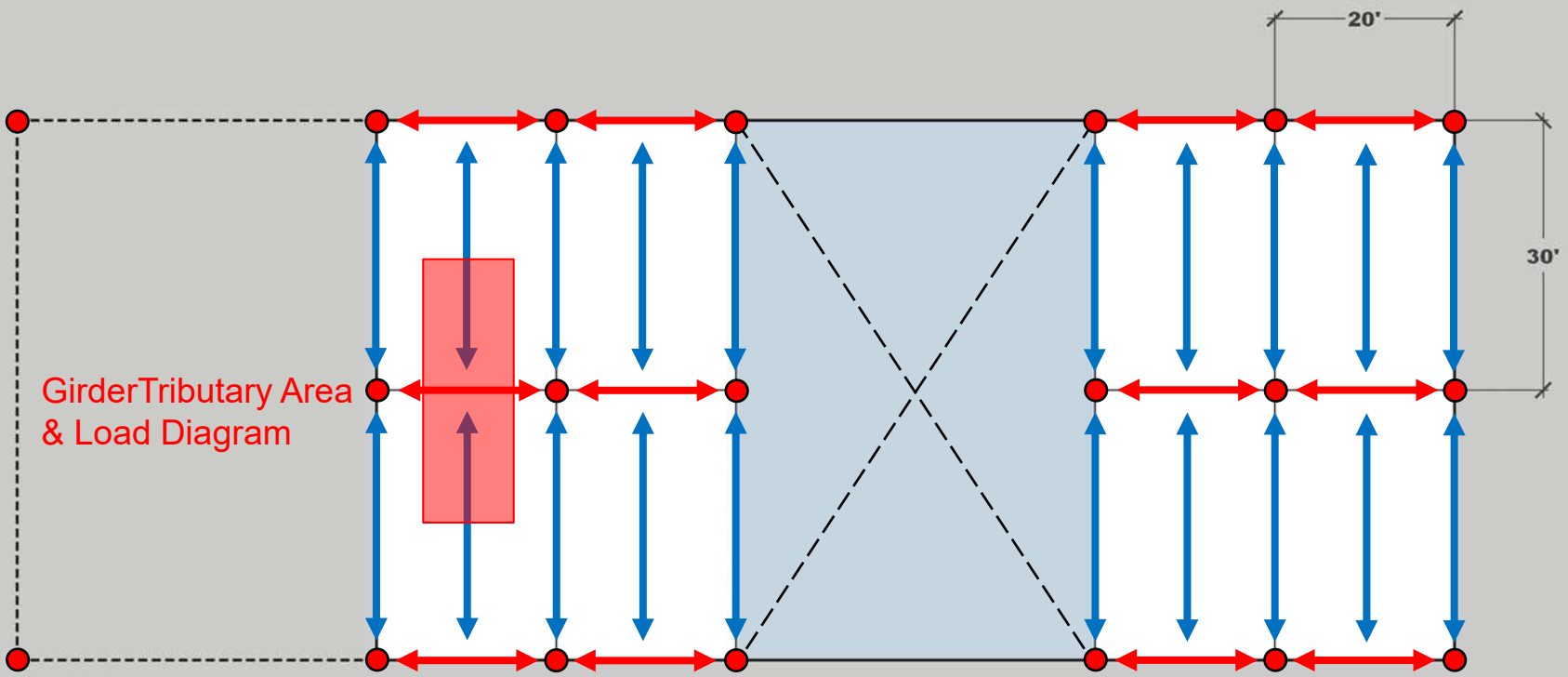
**allow. total load deflection = span/**

**240**

| Load type: | Mmax (ft-kips) | Vmax (kips) | Lightest choice |
|---|---|---|---|
| a) uniformly-distributed load | 308 | 61.5 | *W21×55* |
| b) load at mid-span | 308 | 30.8 | *W21×55* |
| c) loads at third points | 273 | 41.0 | *W21×50* |
| d) loads at quarter points | 308 | 46.1 | *W21×55* |

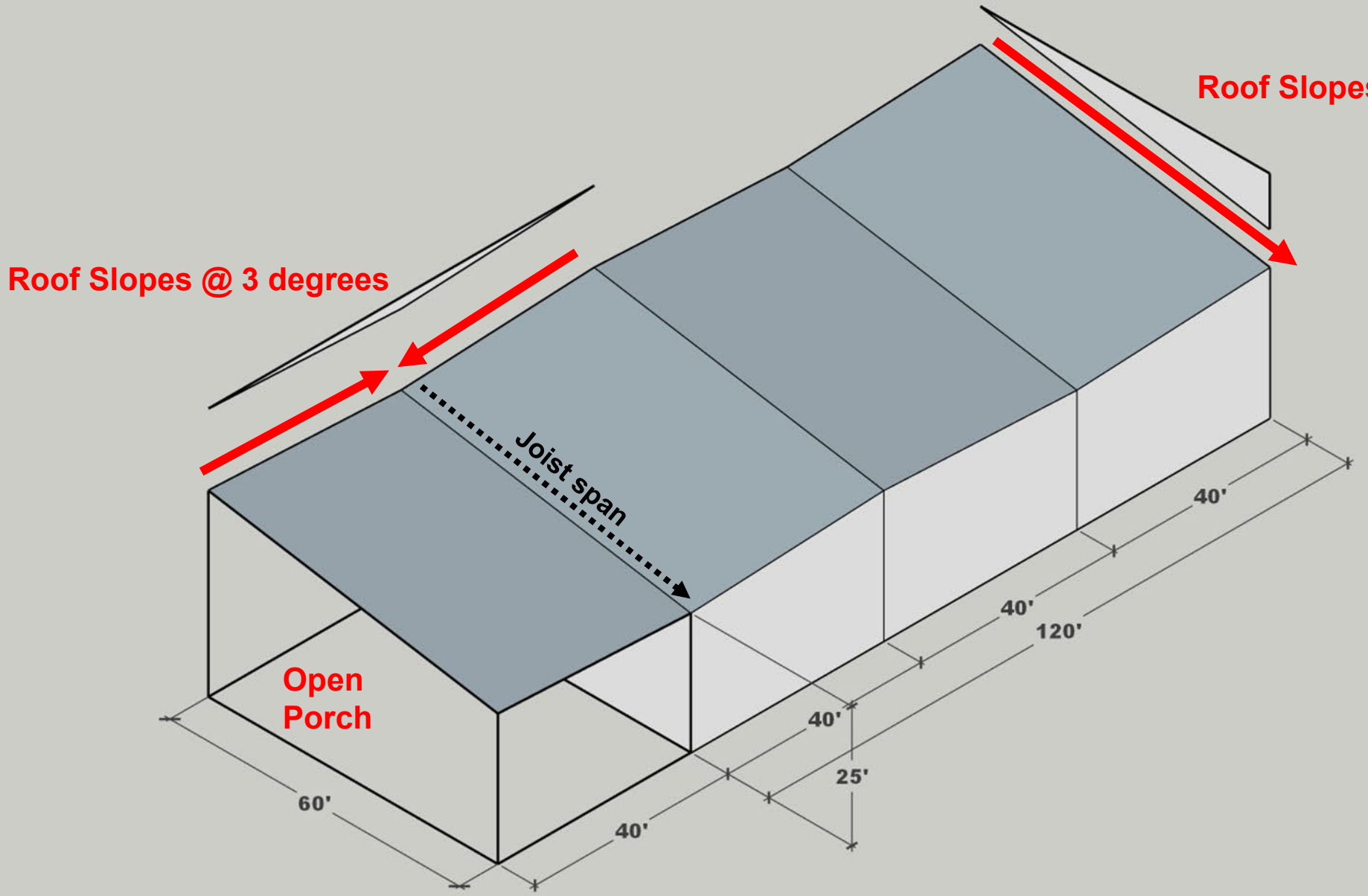
| Design criteria: | | reduction coeff: | live load used: |
|---|---|---|---|
| Use live load reduction? | no | not used | 150 psf |
| Consider live load deflection? | yes | | |
| Consider total load deflection? | no | | |
| Consider shear stress? | yes | | |
| Consider bending stress? | yes | | |
| Lateral support: | continuous | | |

**Girder Size = W21 x 55**

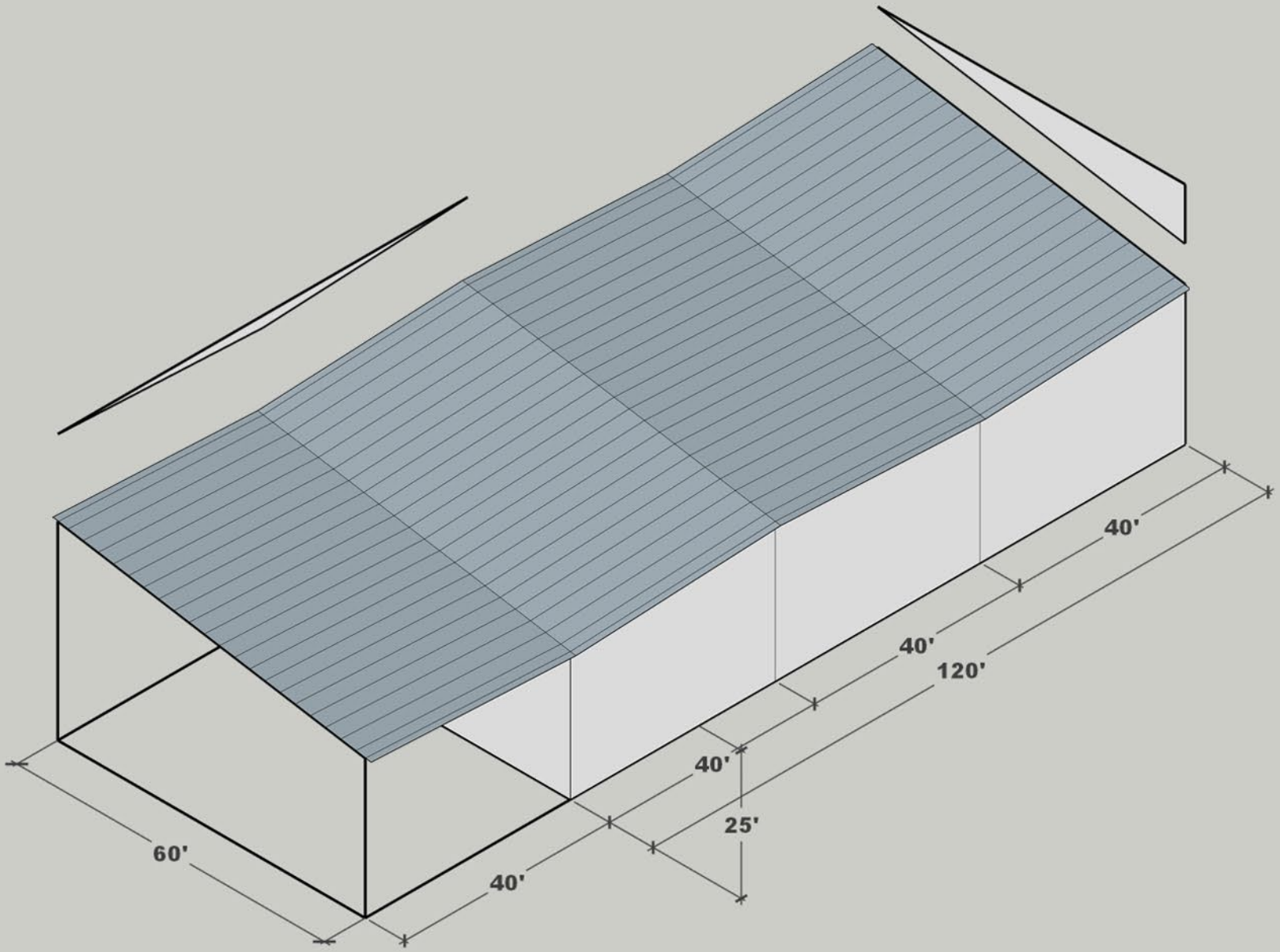



**Girders spaced at 30’ o.c.**
**Girders span 20’ – 0”**
**Girder Load = 205 psf = LL 150 psf + DL 55 psf**
**Girder Size: W21 x 55**

**4060 R** **2nd Floor Plan option selected***…Girder is W21 x 55*

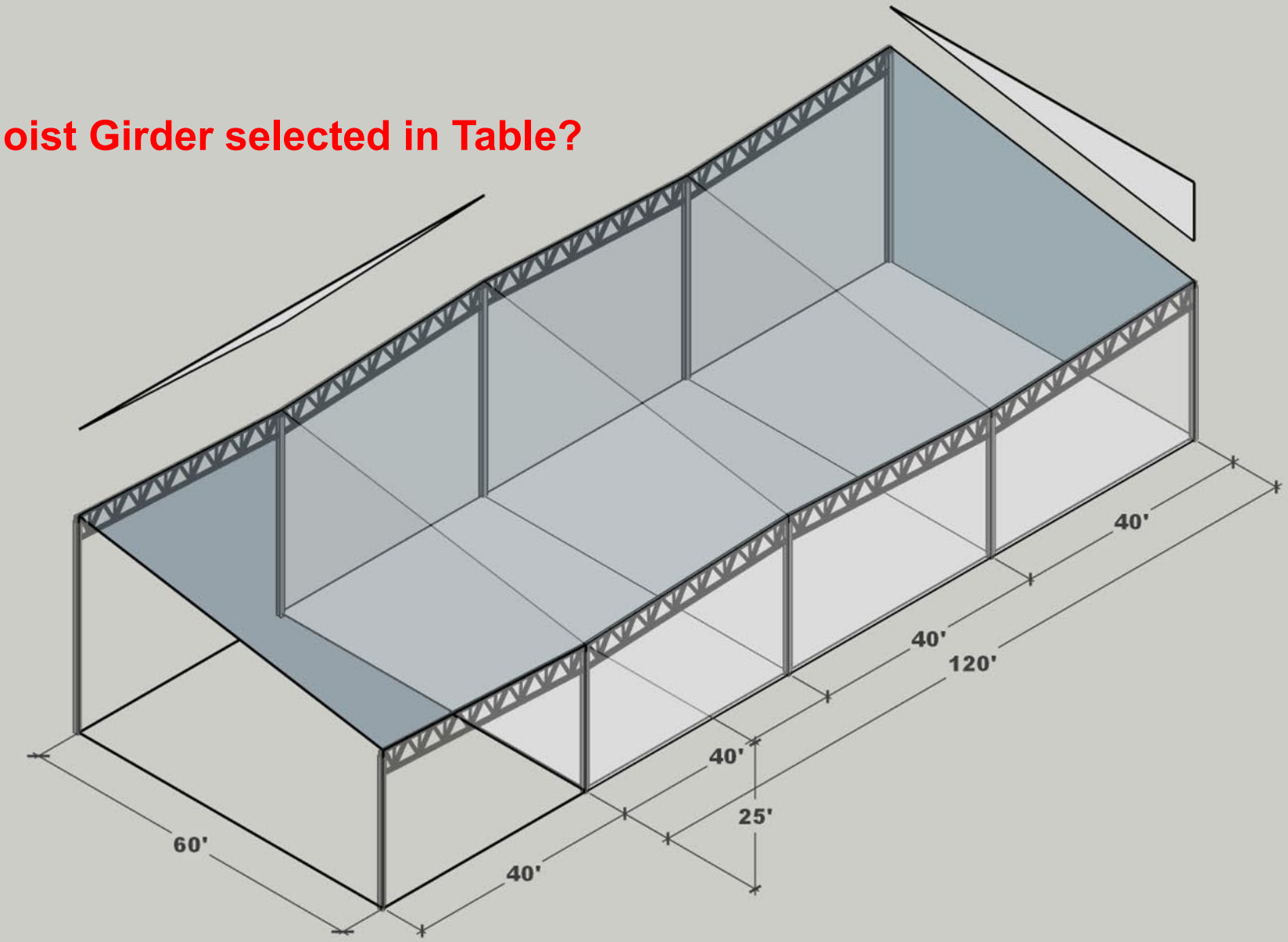



# 4060 C (40’ x 60’ Bays) **(Joists <u>must</u> span 60’)** LIBRARY 100 psf Snow Load



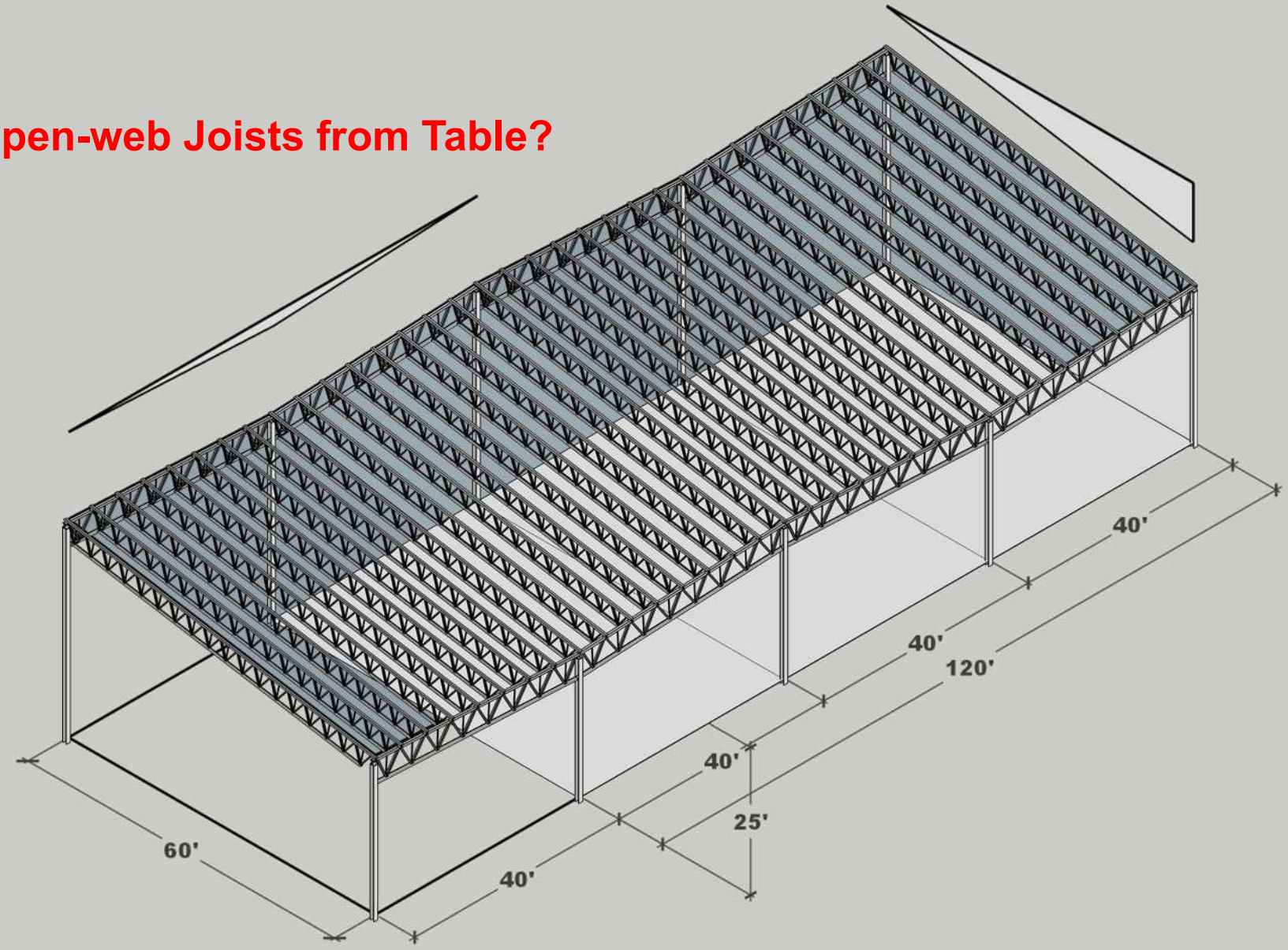

# 4060 C Type-B Roof Deck (1.5” x 36” wide panels)

**Label specific Joist Girder selected in Table?**

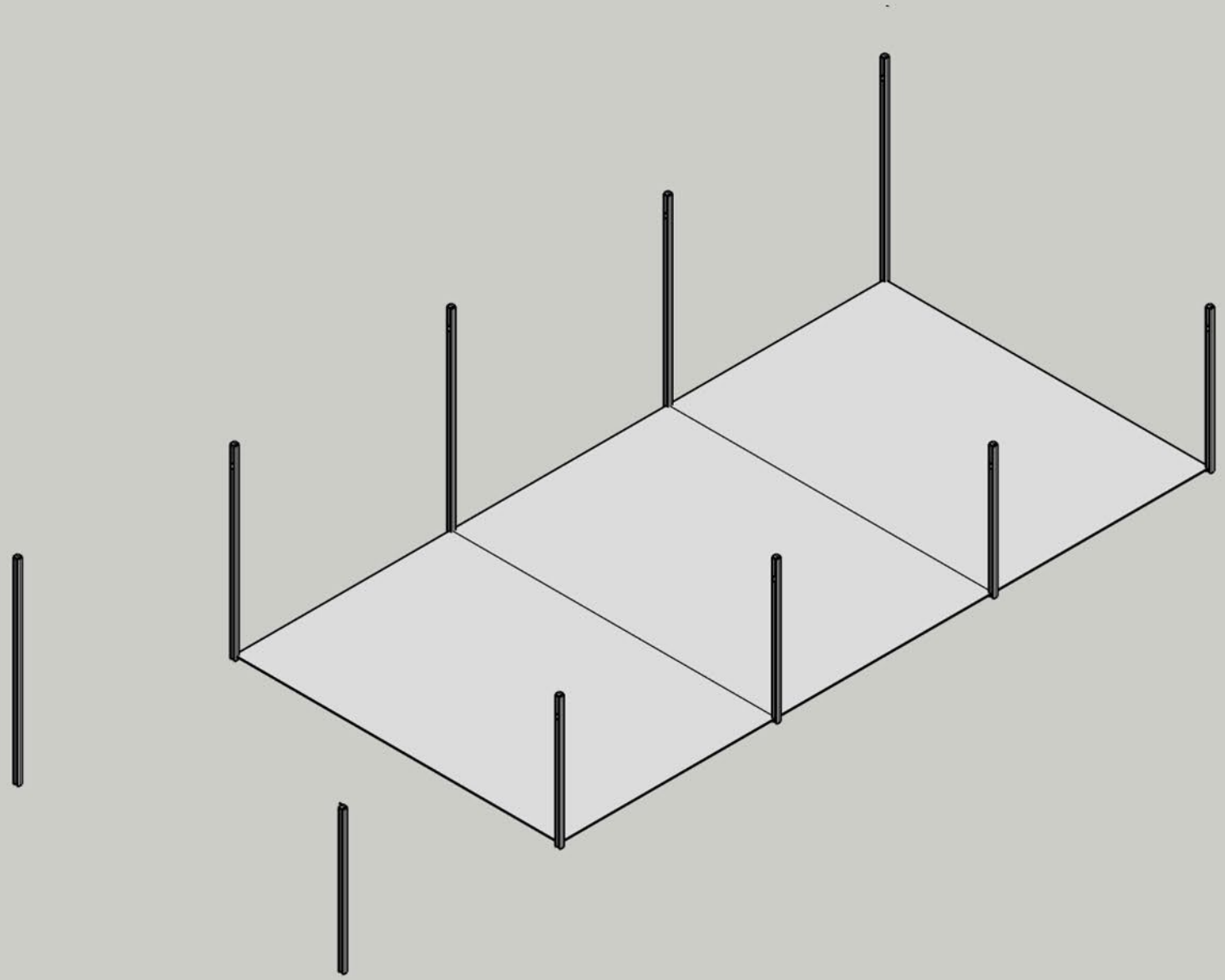



# 4060 C **Roof Columns and Joist Girders** (8 Panels @ 5’ wide = 40’ span)

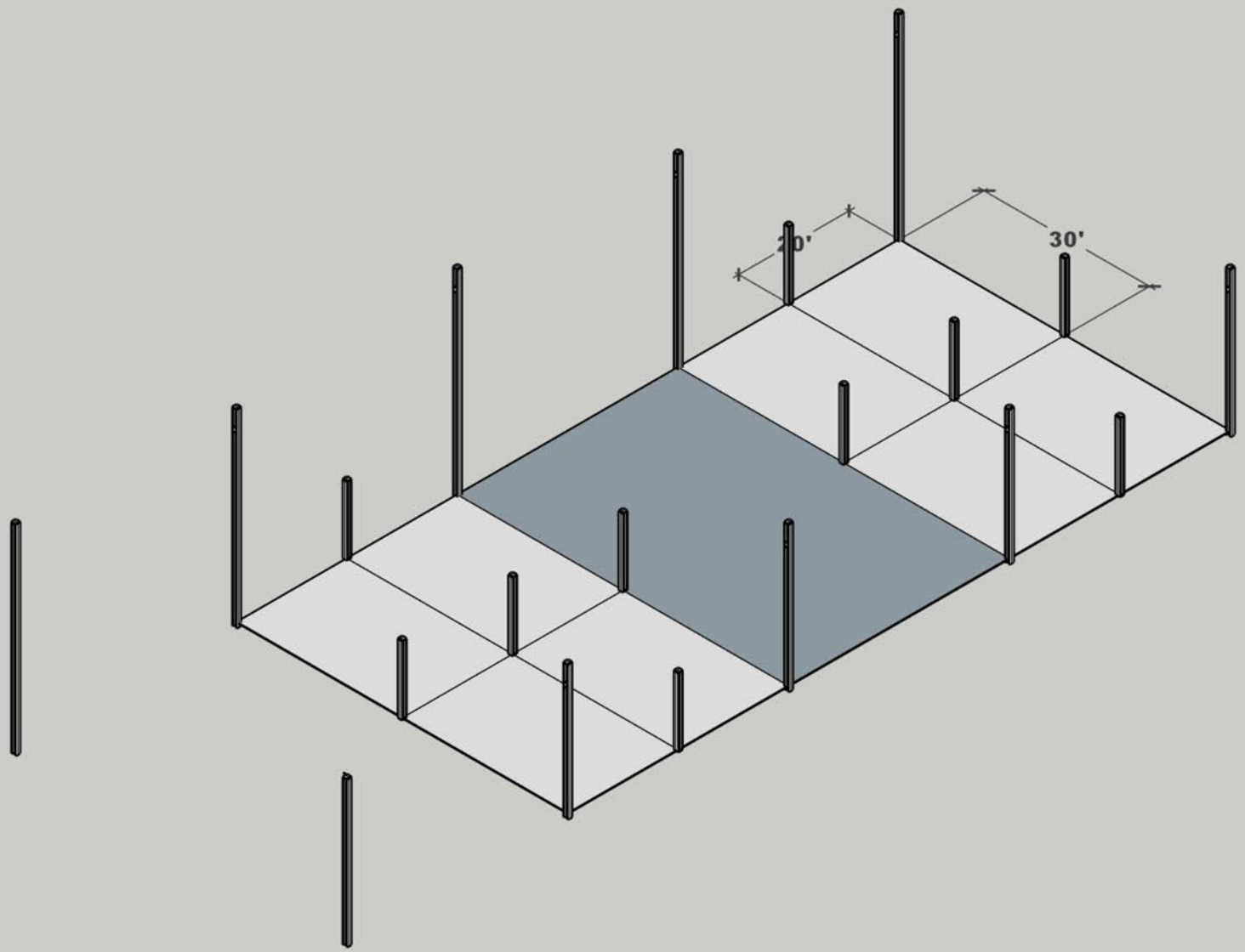
**Label specific Open-web Joists from Table?**




**4060 C Roof Columns and Joists** (60’ span, Joists spaced at 5’ o.c. to support Deck)

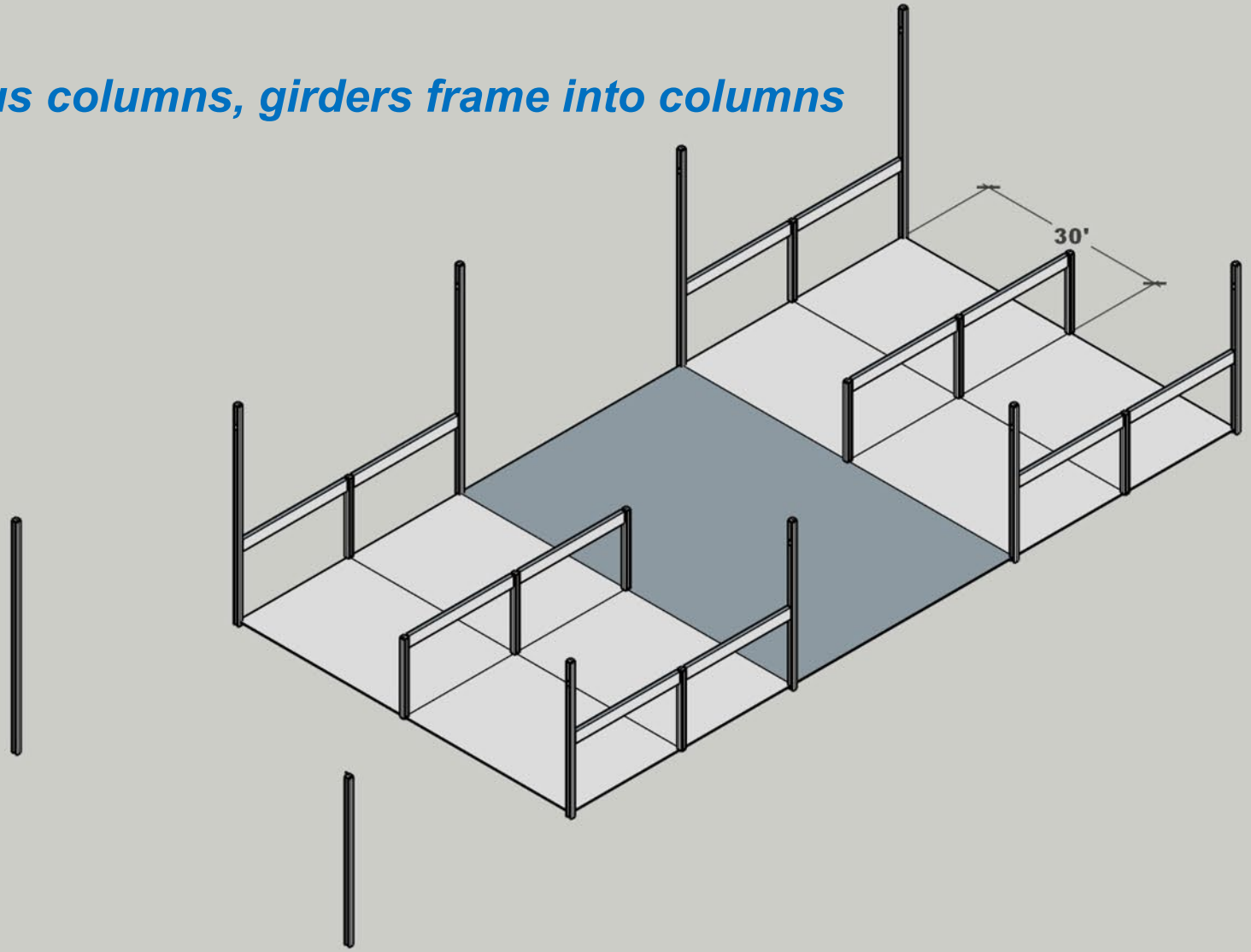


# 4060 C Roof Columns

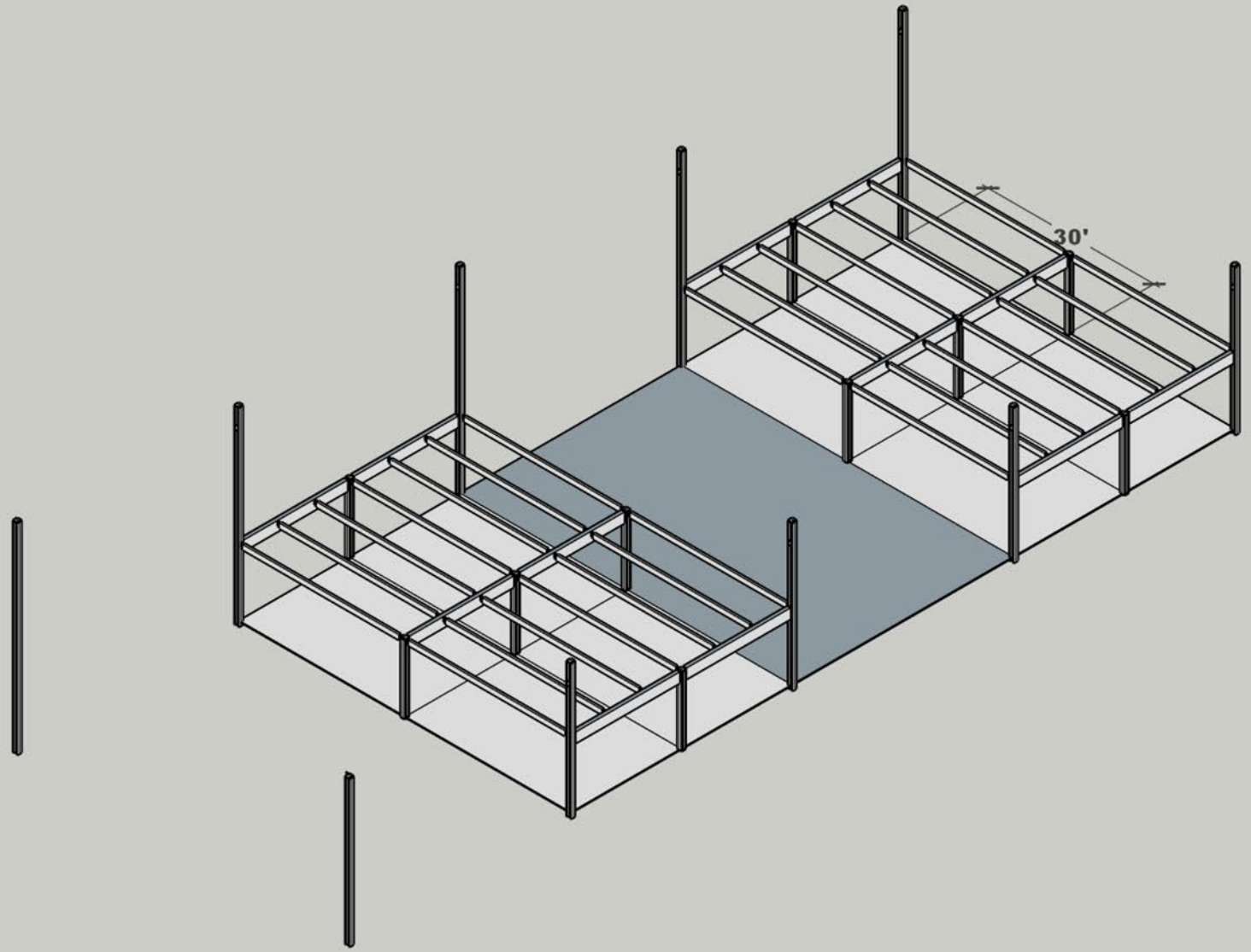



**4060 C** **Roof Columns and** **R** **Floor Columns** (20’ x 30’ Structural Floor Bays) 1:1.5 ratio

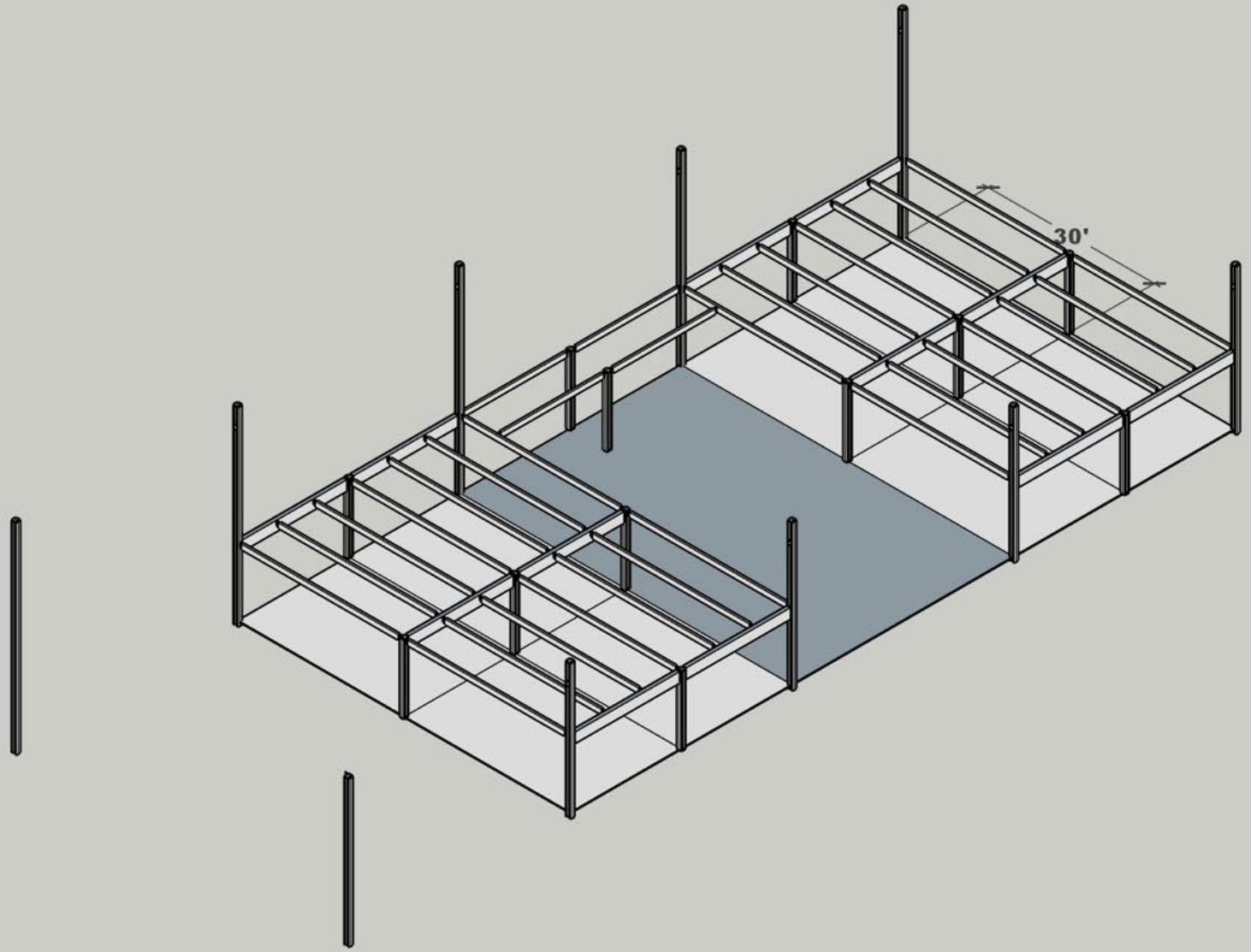
*…model continuous columns, girders frame into columns*



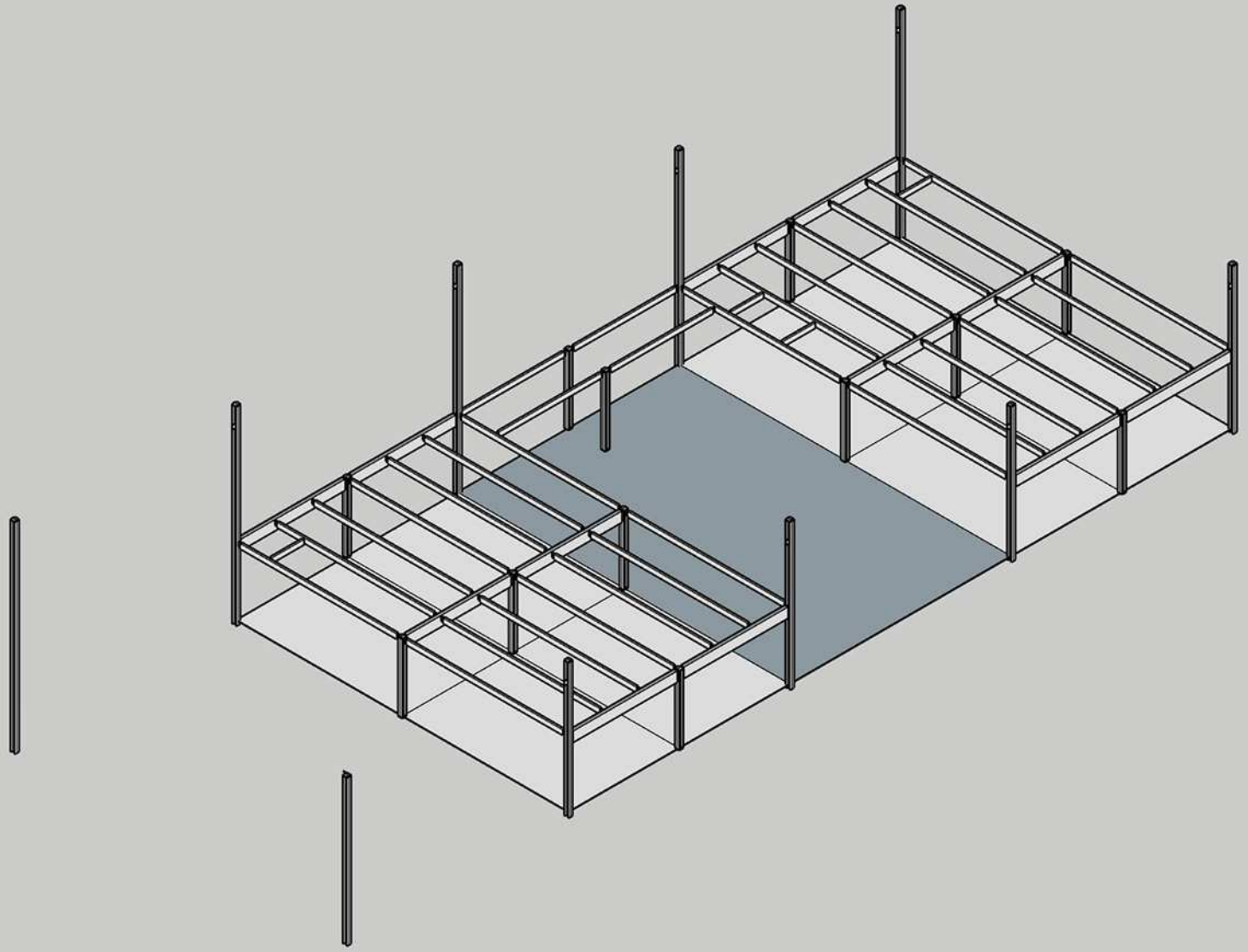

**4060 R Columns and Wide-Flange Girders W21 x 55** (Girders span 20’ to columns)



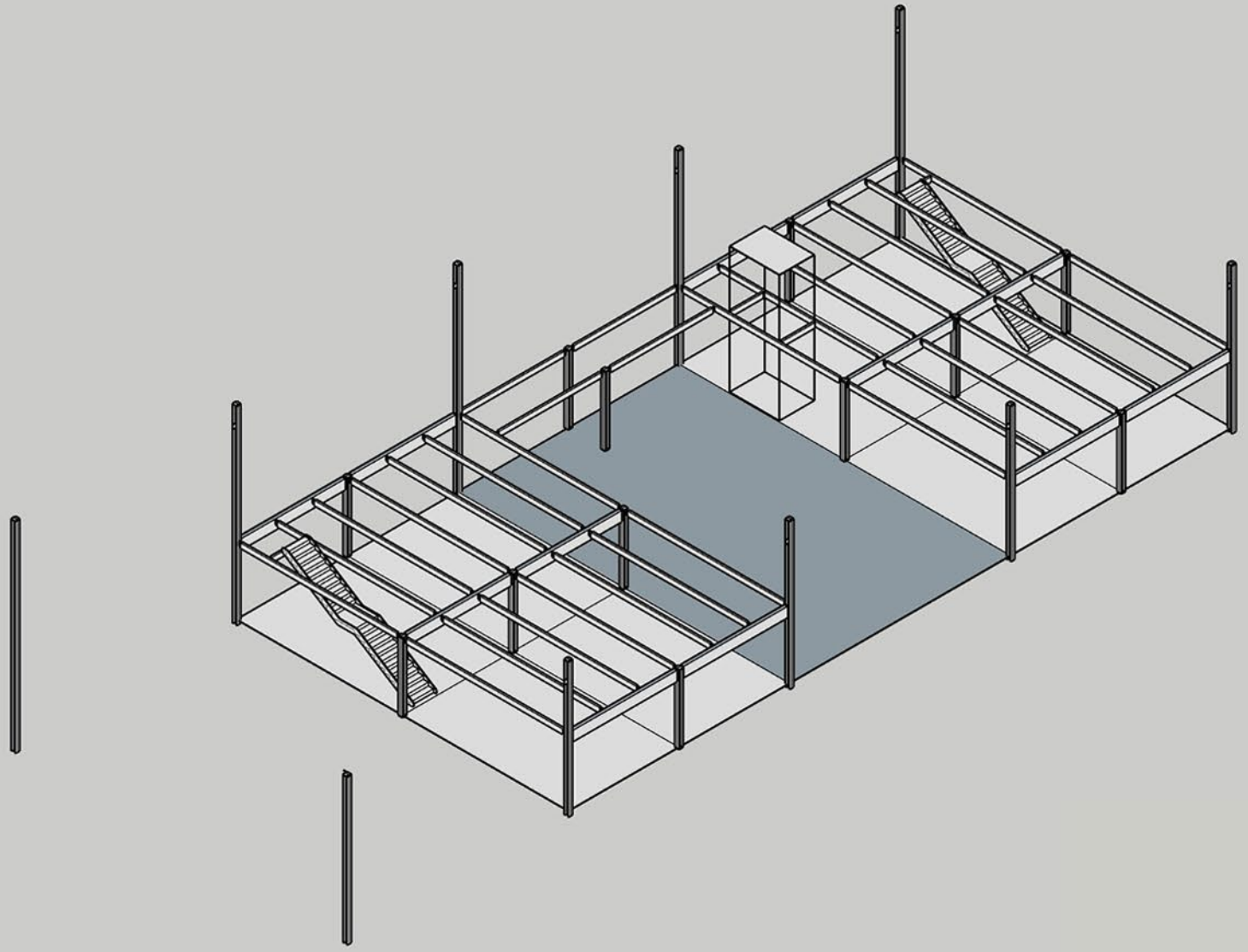

**4060 R Columns, Girders and Beams W21 x 44** (Beams span 30’ to Girders (and Columns))



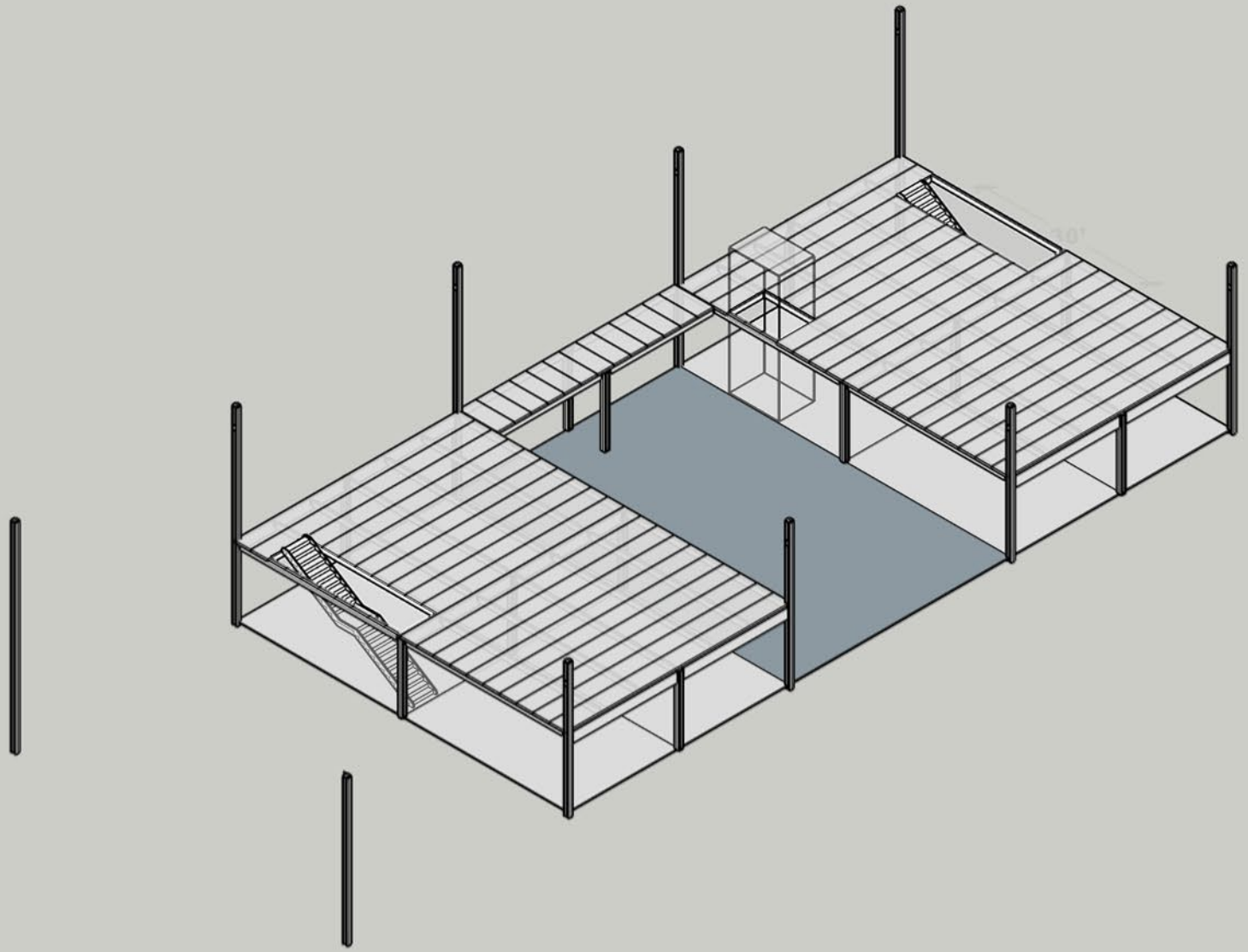

# 4060 R Add Columns and Beams for bridge



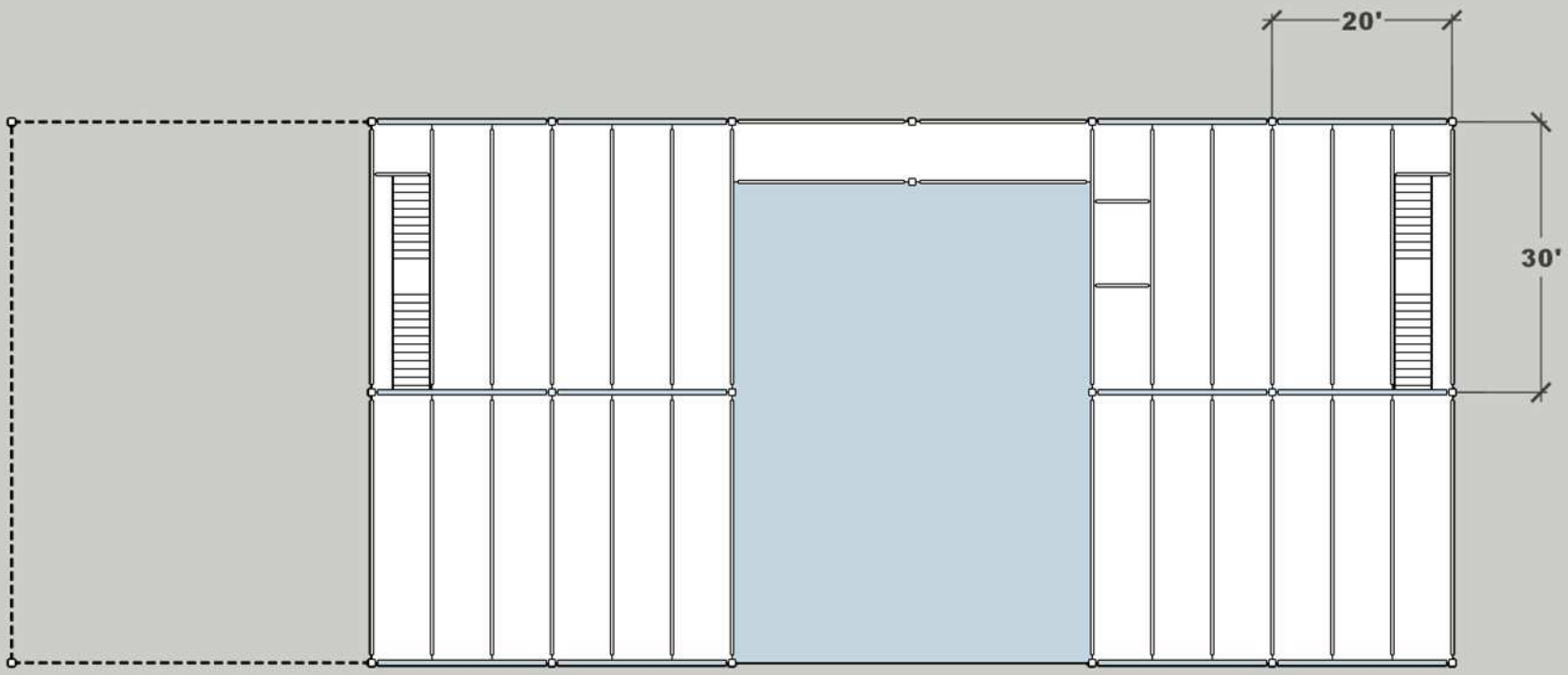
**4060 R** **Add structural opening for stairs and opening for elevator (header beams)**



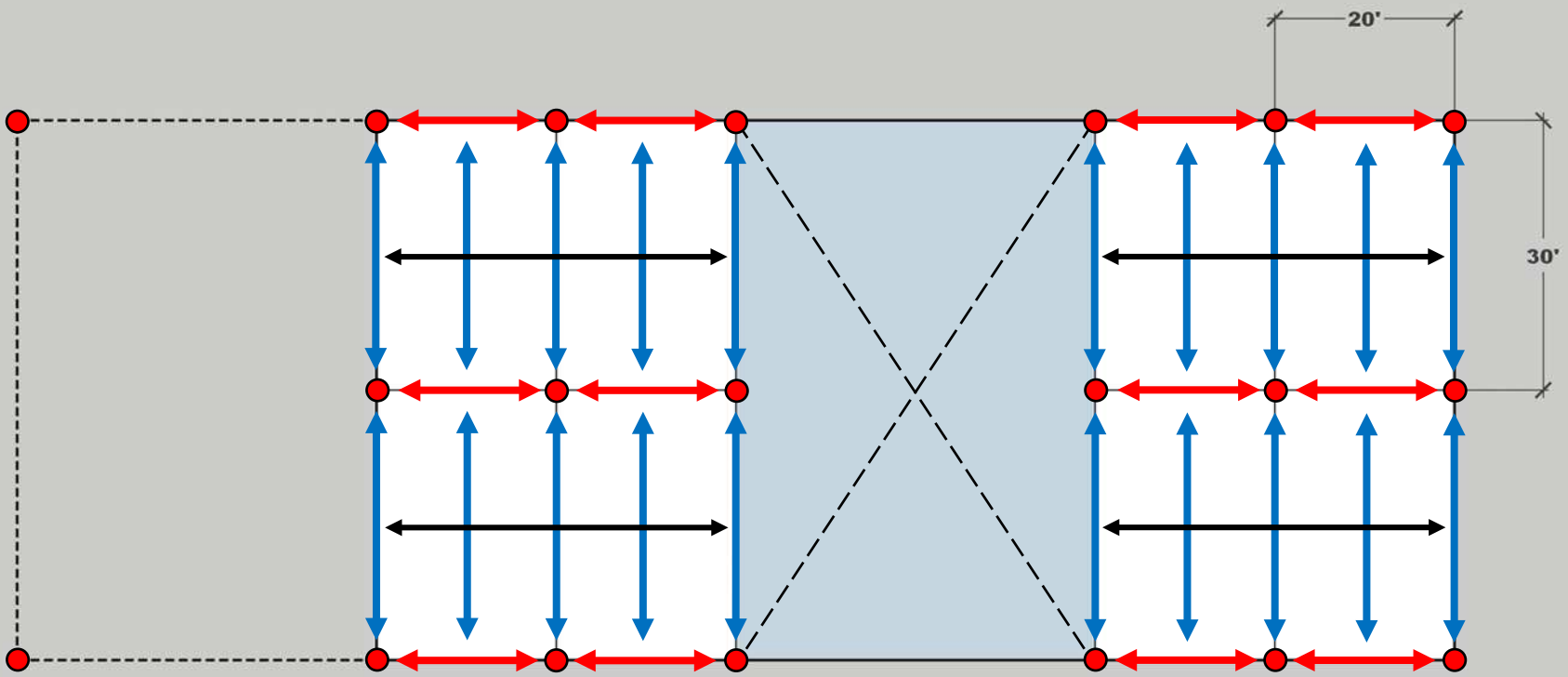
# 4060 R Add structural Stairs and elevator



**4060 R** Composite Deck spans 6.67’ over Beams (Deck corrugation in 3’ width)




# 4060 R Plan @ Second Floor*…Label and Dimension*



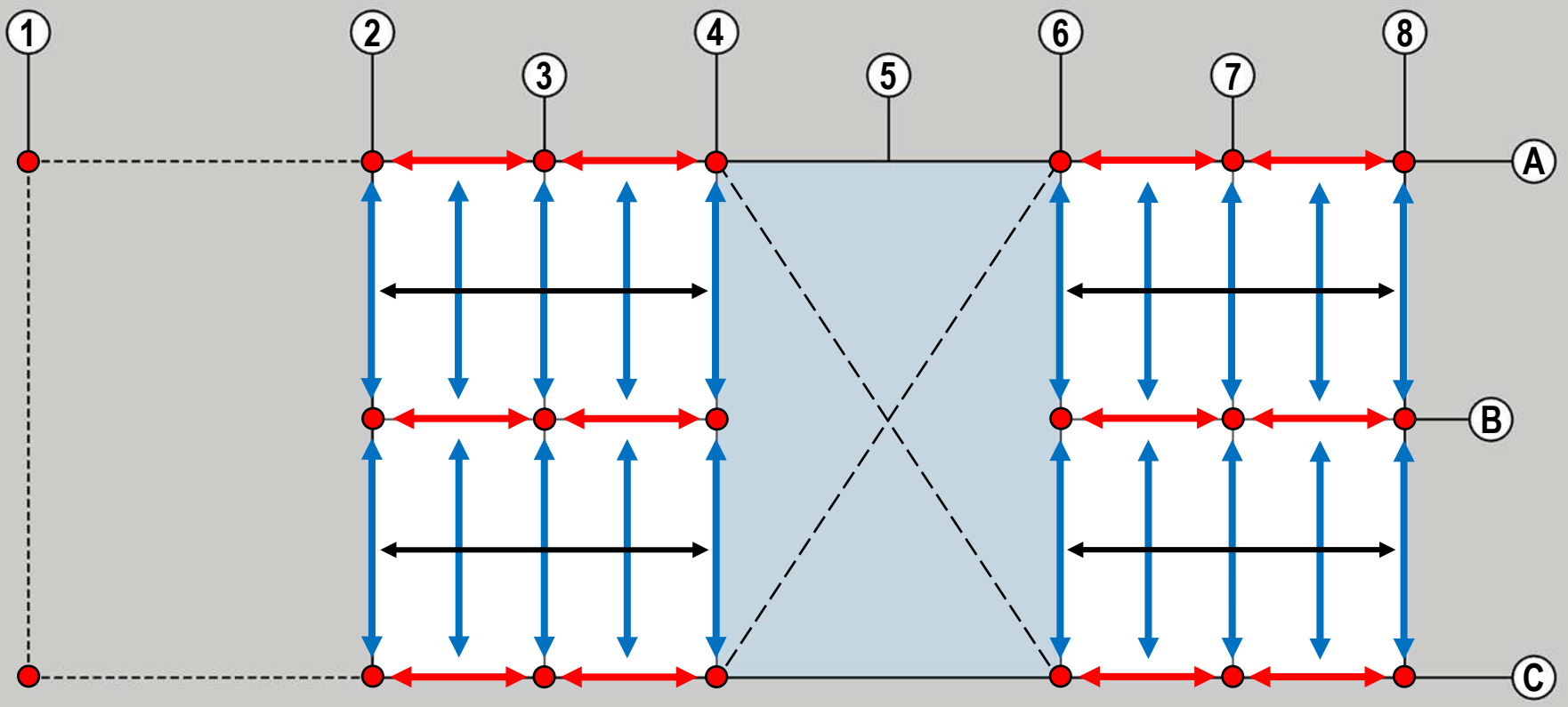

Deck span direction

Beam span direction and spacing

Girder span direction and spacing

Column locations

**EXAMPLE: Framing Plan Diagram of floor structure**




**Deck span direction**
**Beam span direction and spacing**
**Girder span direction and spacing**
**Column locations**

## 2ND FLOOR FRAMING PLAN DIAGRAM…*with grid lines added*

# STEEL FLOOR

**SECONDARY HORIZONTAL STRUCTURE**

## BEAM SPACING

Beams are spaced at 6.67’ o.c. to optimize the floor deck to span as close to 7’ as possible, while conforming to the 20’ x 30’ structural bay for the floor system. The 20 x 30 bay is the preferred bay ratio of 1.5 to 1, with Beams spanning in the 30’ direction and Girders spanning 20’.

## TRIBUTARY LOAD ON BEAM A

Composite Deck Floor Load: 198 psf
Deck span and Beam spacing: 6.67’
Uniformly Distributed Tributary Load on Beam calculation: 198 psf (6.67’) = 1,321 plf
**LOAD ON BEAM A: 1,321 plf**

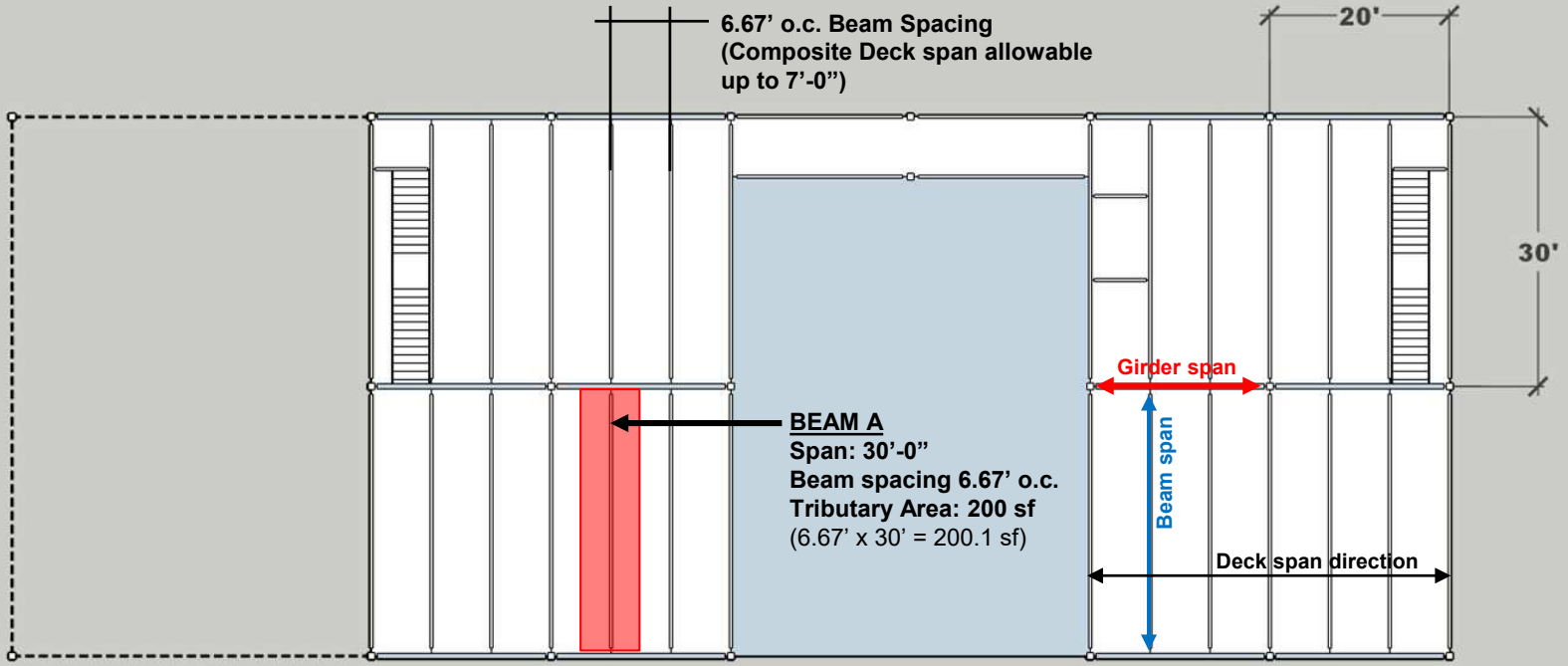
## SIZING OF BEAM A

Sizing based on values entered and results indicated using the following online structural Steel Beam Calculator:
https://jonochshorn.com/scholarship/calculators-st/example8.6/index.html

Results indicated a wide-flange beam W18 x 35 is the lightest section to meet the load and beam span criteria. This beam is 17.7 inches deep and weighs 35 plf.

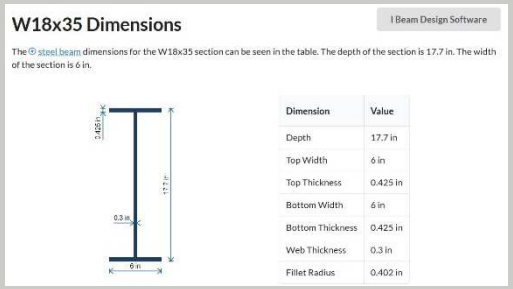
**SELECTION FOR BEAM A: W18 X 35**

**FORMAT!!!**

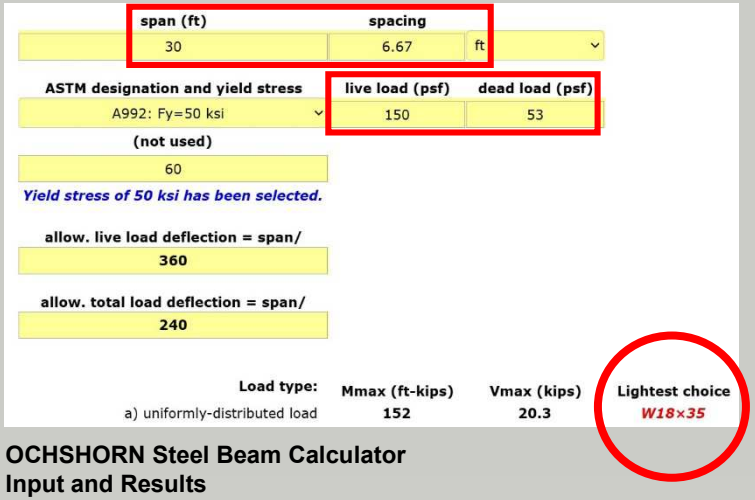



## BEAM A

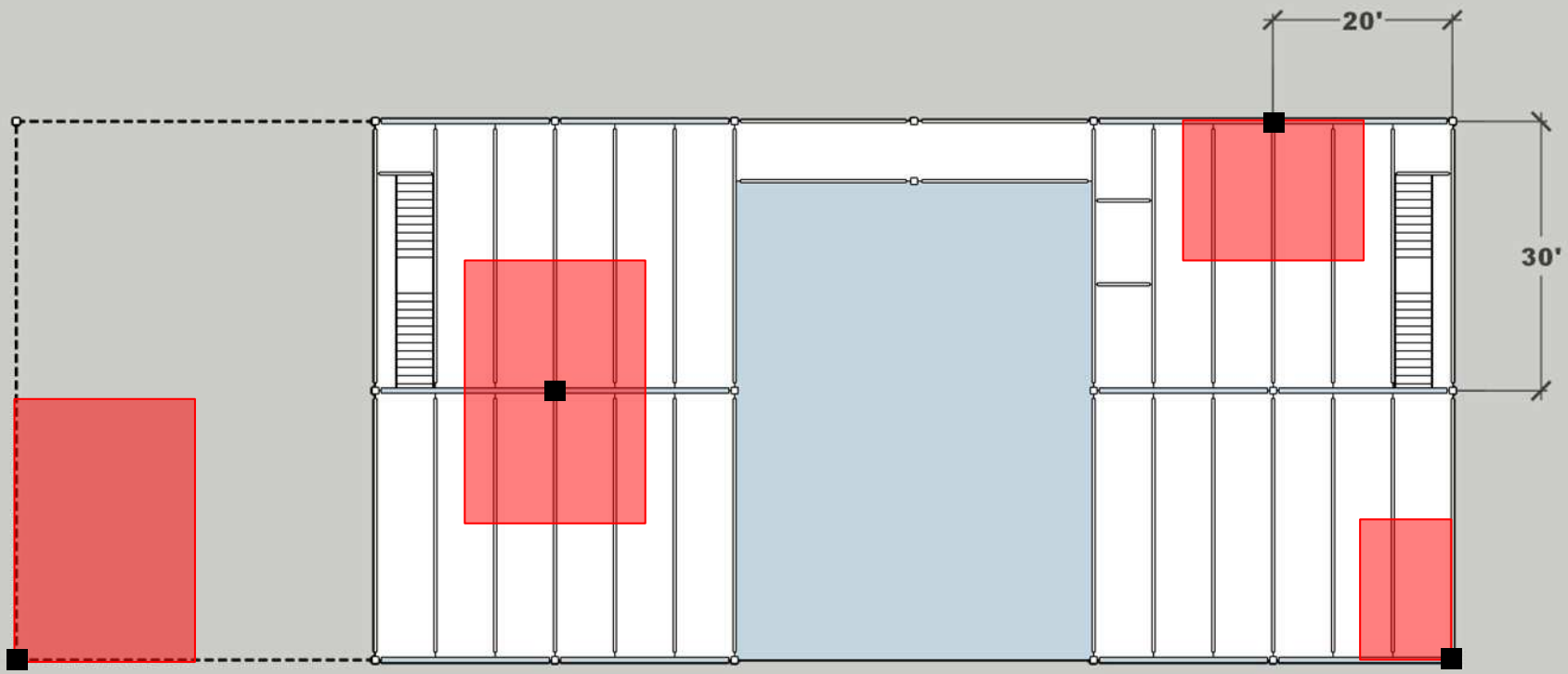
**Total Tributary Load: 39.62 kips**
(198 psf x 200.1 sf)
**Uniform Distributed Load: 1,321 plf**
(198 psf x 6.67’ = 1,321 plf)



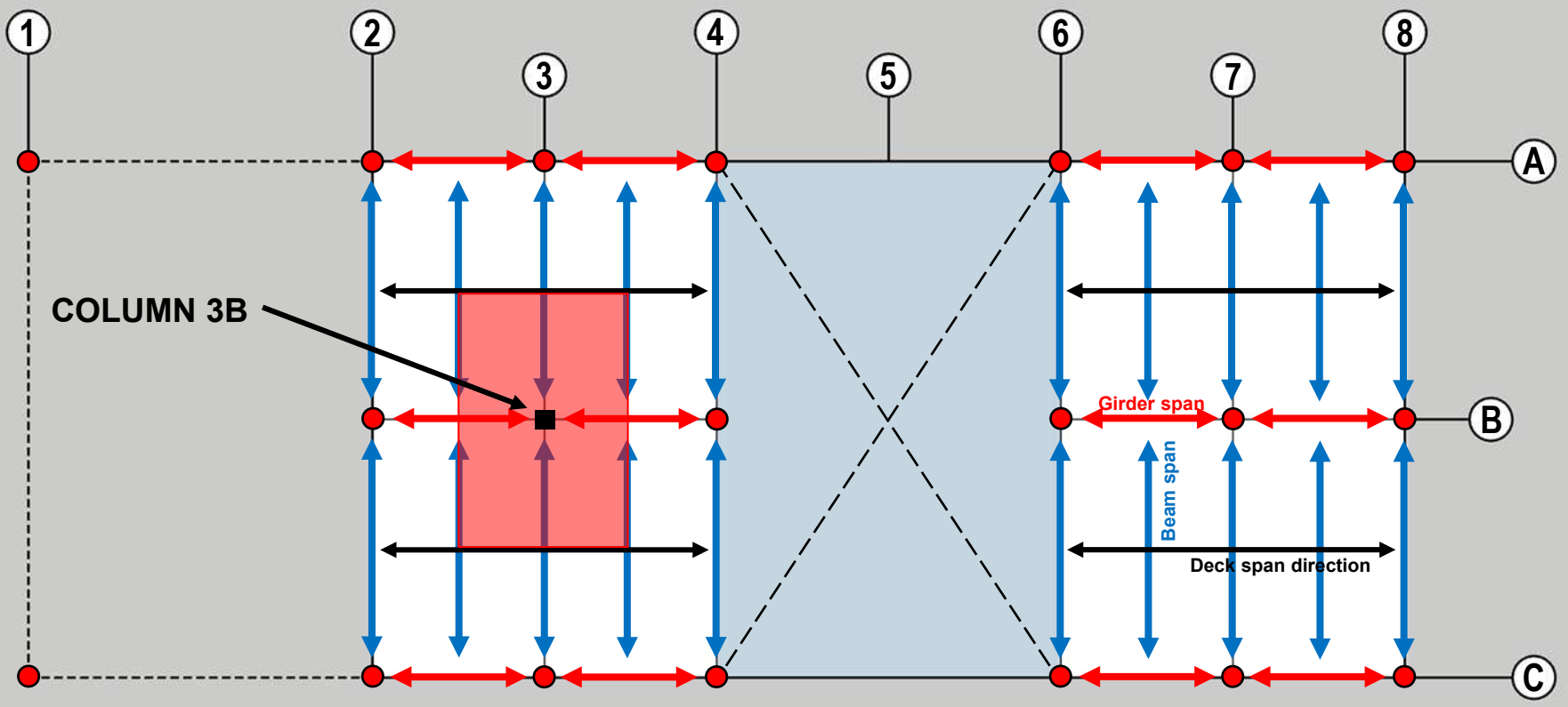

**W18X35 size**




**OCHSHORN Steel Beam Calculator Input and Results**

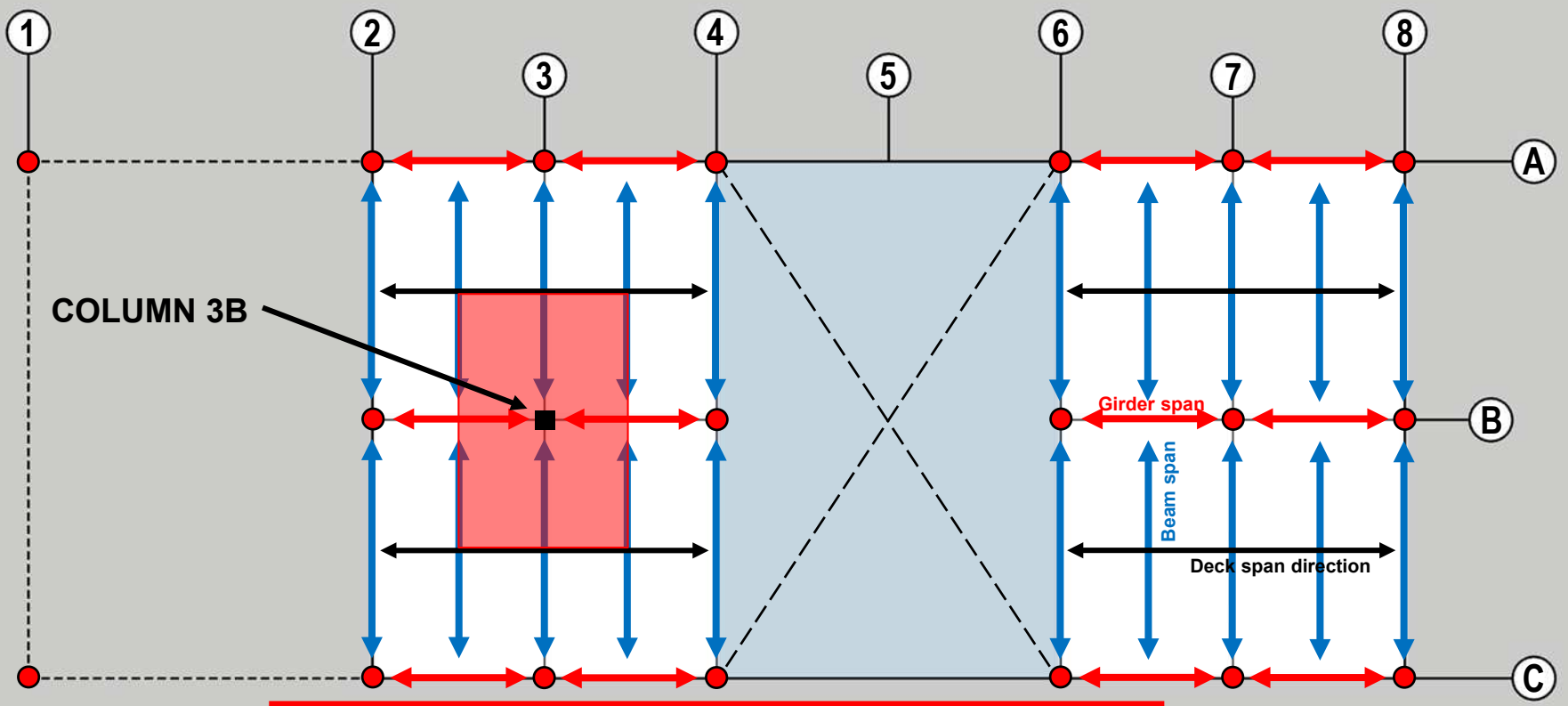



**4060 C** Framing Plan @ Second Floor *…Find Load and size of Columns?*



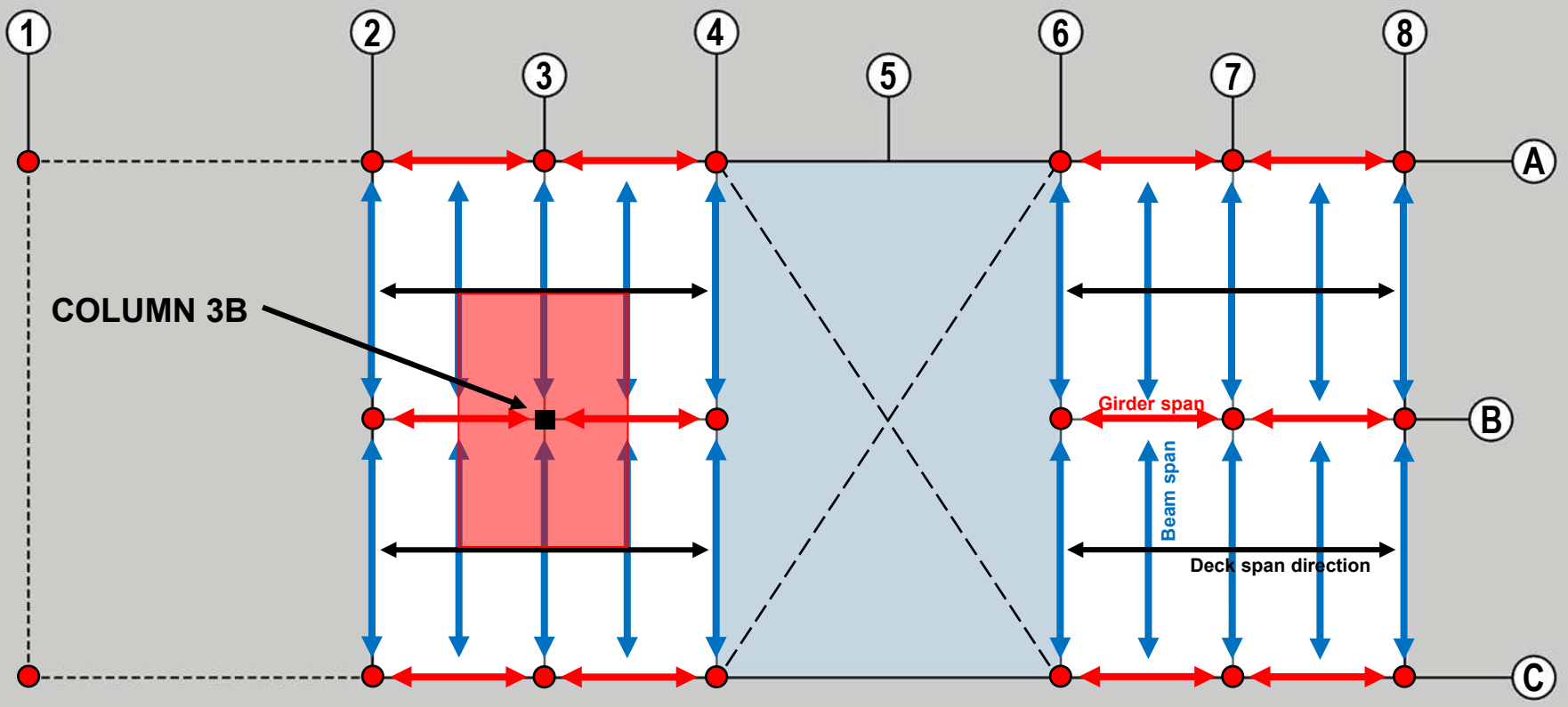

*…add grid lines to your plan diagrams!*
*…then each column will have a unique number & letter code*

**COLUMN 3B***…add grid lines to diagram…find the axial load on column 3B*…*(floor load only)*




**COLUMN 3B** (Floor-Load only)
**Floor Deck Tributary Area = 600 sf** = 20' x 30'
**Floor Deck Tributary Load = 118.8 kips** = 600 sf (198 psf)

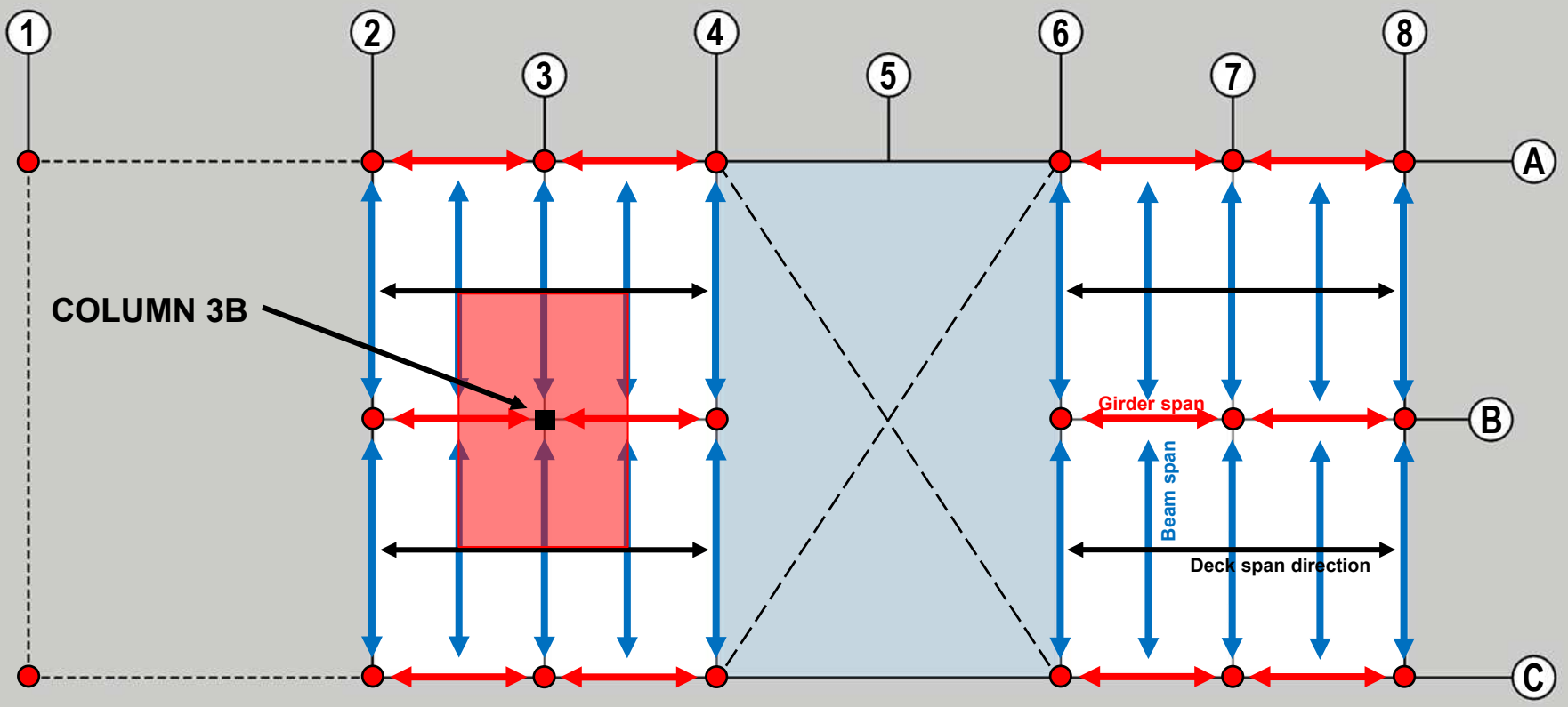
## COLUMN 3B*…find the Floor Deck Area & Load*




**<u>COLUMN 3B</u> (Floor-Load only)**
**Floor Deck Tributary Area = 600 sf** = 20’ x 30’
**Floor Deck Tributary Load = 118.8 kips** = 600 sf (198 psf)

**Beam Self-weight = 1.32 kips** = 44 plf (30’ length of W21 x 44)
**Girder Self-weight = 1.1 kips** = 55 plf (20’ length of W21 x 55)

## COLUMN 3B*…find the self-weight of Beam & Girder*

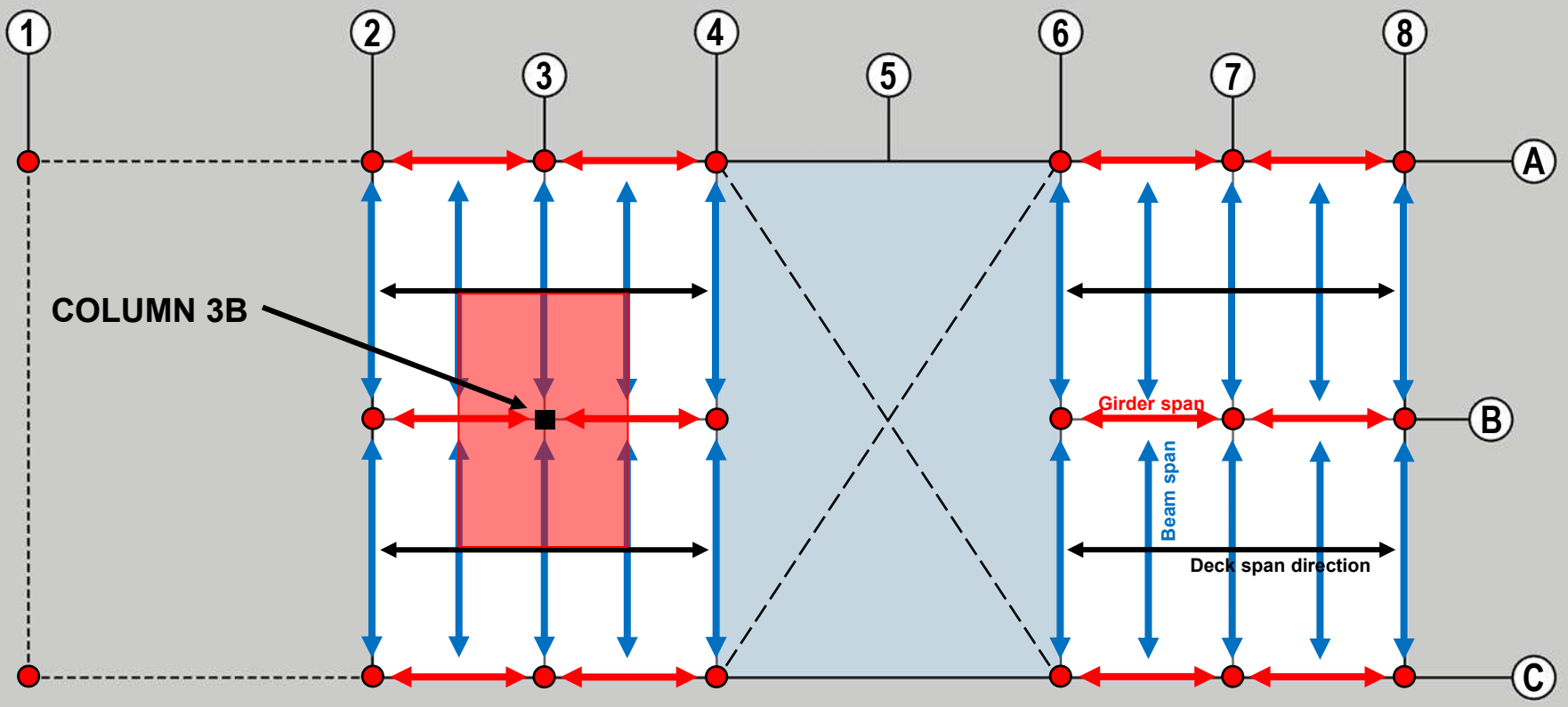



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**Beam Self-weight = 1.32 kips** = 44 plf (30' length of W21 x 44)
**Girder Self-weight = 1.1 kips** = 55 plf (20' length of W21 x 55)

**Total Tributary Load on Column 3B = 121.22 kips**
(118.8 kips + 1.32 kips + 1.1 kips)

**COLUMN 3B***…add the loads of Floor Deck, Beam & Girder for the axial load on the column*




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**Floor Deck Tributary Load = 118.8 kips** = 600 sf (198 psf)
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**Girder Self-weight = 1.1 kips** = 55 plf (20' length of W21 x 55)
**Total Tributary Load on Column 3B = 121.22 kips**
(118.8 kips + 1.32 kips + 1.1 kips)

**121.22 kips = Column 3B Axial Load**
12.25' = Column 3B Height
ASTM A992 Steel (50 ksi Yield Strength)

**COLUMN 3B***…highlight the axial load on the column*

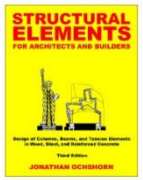
*…find the column size of Column 3B*

**COLUMN 3B**

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## Steel wide-flange column design calculator

Jonathan Ochshorn

© 2016 Jonathan Ochshorn.

| | | |
|---|---|---|
| Steel type | A992 | |
| Height | 12.25 | feet |
| K | 1 | |
| Axial load | 121.22 | kips |
| Limit KL/r to 200 | | |

**Five lightest choices:**

| | |
|---|---|
| 1 | ***W8x28*** |
| 2 | W12x30 |
| 3 | W14x30 |
| 4 | W8x31 |
| 5 | W10x33 |

**Steel Type: A992**
This type has ultimate yield strength of 50 ksi

**Height: The unbraced height of the column**
Also know as the effective length

**K: The effective length factor**
This factor influences the unbraced height and the connection type at the top and bottom of the column.
**K = 1** for a steel system with lateral-load resistant braced frames and roof/floor diaphragms.

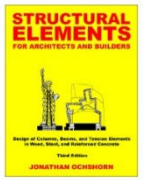
**Axial Load: The load applied to the column**

**COLUMN 3B*…use this calculator***

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| | | |
|---|---|---|
| Steel type | A992 | |
| Height | 12.25 | feet |
| K | 1 | |
| Axial load | 121.22 | kips |
| Limit KL/r to 200 | | |

**Five lightest choices:**

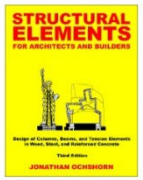
| | |
|---|---|
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| 3 | W14x30 |
| 4 | W8x31 |
| 5 | W10x33 |

**COLUMN 3B*…use this calculator***

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**Axial Load: The load applied to the column**

| | | |
|---|---|---|
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| Height | 12.25 | feet |
| K | 1 | |
| Axial load | 121.22 | kips |
| Limit KL/r to 200 | | |

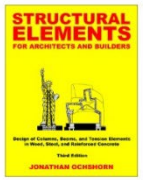
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**Axial Load: The load applied to the column**

| | | |
|---|---|---|
| Steel type | A992 | |
| Height | 12.25 | feet |
| K | 1 | |
| Axial load | 121.22 | kips |
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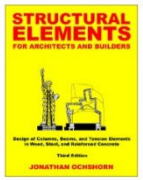
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**K: The effective length factor**
This factor influences the unbraced height and the connection type at the top and bottom of the column.
**K = 1** for a steel system with lateral-load resistant braced frames and roof/floor diaphragms.

**Axial Load: The load applied to the column**
From your calculations of specific column

| | | |
|---|---|---|
| Steel type | A992 | |
| Height | 12.25 | feet |
| K | 1 | |
| Axial load | 121.22 | kips |
| Limit KL/r to 200 | | |

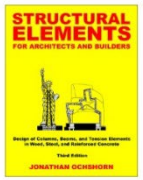
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Jonathan Ochshorn

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| | | |
|---|---|---|
| Steel type | A992 | |
| Height | 12.25 | feet |
| K | 1 | |
| Axial load | 121.22 | kips |
| Limit KL/r to 200 | | |

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| | |
|---|---|
| 1 | ***W8x28*** |
| 2 | W12x30 |
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| 4 | W8x31 |
| 5 | W10x33 |

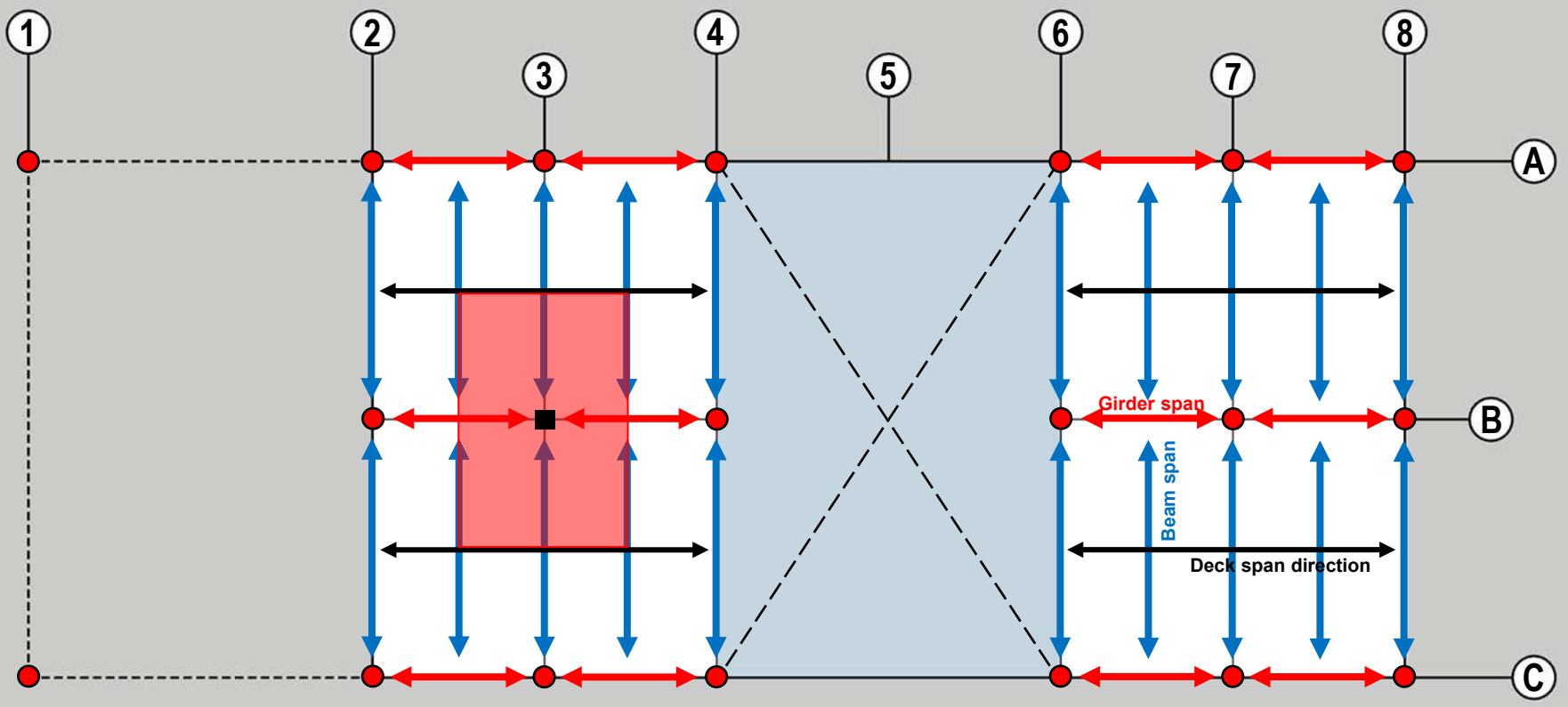
**Steel Type: A992**
This type has ultimate yield strength of 50 ksi

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Also know as the effective length

**K: The effective length factor**
This factor influences the unbraced height and the connection type at the top and bottom of the column.
**K = 1** for a steel system with lateral-load resistant braced frames and roof/floor diaphragms.

**Axial Load: The load applied to the column**

**Column Size = W8 x 28**




**COLUMN 3B (Floor-Load only)**

**Floor Deck Tributary Area = 600 sf** = 20' x 30'
**Floor Deck Tributary Load = 118.8 kips** = 600 sf (198 psf)
**Beam Self-weight = 1.32 kips** = 44 plf (30' length of W21 x 44)
**Girder Self-weight = 1.1 kips** = 55 plf (20' length of W21 x 55)
**Total Tributary Load on Column 3B = 121.22 kips**
(118.8 kips + 1.32 kips + 1.1 kips)

**121.22 kips = Column 3B Axial Load**
12.25' = Column 3B Height
ASTM A992 Steel (50 ksi Yield Strength)

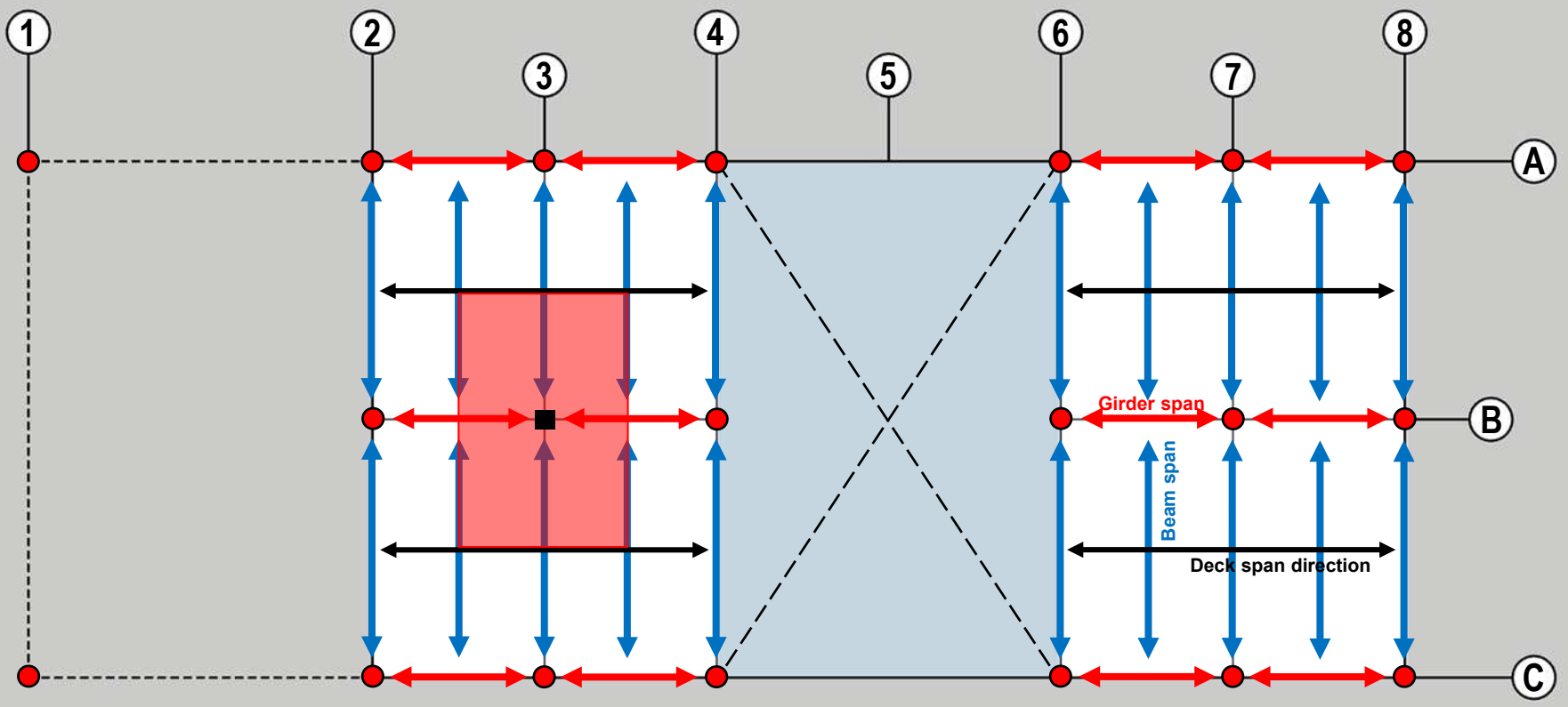
**W8 x 28: Column 3B**

| | | |
|---|---|---|
| Steel type | A992 | |
| Height | 12.25 | feet |
| K | 1 | |
| Axial load | 121.22 | kips |
| Limit KL/r to 200 | | |

**Five lightest choices:**

| | |
|---|---|
| 1 | W8x28 |

# COLUMN 3B




**<u>COLUMN 3B</u> (Floor-Load only)**
**Floor Deck Tributary Area = 600 sf** = 20’ x 30’
**Floor Deck Tributary Load = 118.8 kips** = 600 sf (198 psf)
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(118.8 kips + 1.32 kips + 1.1 kips)

**121.22 kips = Column 3B Axial Load**
12.25’ = Column 3B Height
<u>ASTM A992 Steel (50 ksi Yield Strength)</u>
**W8 x 28: Column 3B**

# COLUMN 3B and FOUNDATION 3B*…what size is the foundation?*

## PROMPT

Select the size of the area and the thickness of a spread footing to support an interior steel column.
The axial load on the column is 121.22 kips.
The column is a W8 x 28 section.
The depth of the footing must be 42 inches below grade.
The column is supported by a 20-inch x 20-inch concrete pedestal resting on the footing.
Assume the concrete bearing capacity is 3,000 psf.
The allowable bearing pressure of the soil is 1,500 psf.
Use IBC basic load combinations.

## Find Foundation using Google Gemini

### 1. Footing Plan Area Selection

To determine the required footing base area ($A$), allowable stress design (ASD) is used under the service axial load $P = 121.22$ kips:

$$A_{\text{req}} = \frac{P}{q_{\text{allow}}} = \frac{121.22 \text{ kips}}{1.5 \text{ ksf}} = 80.81 \text{ ft}^2$$

For a square footing of width $B$:

$$B = \sqrt{A_{\text{req}}} = \sqrt{80.81 \text{ ft}^2} \approx 8.99 \text{ ft}$$

- **Selected Footing Size:** $9.0 \text{ ft} \times 9.0 \text{ ft}$ $(108 \text{ in.} \times 108 \text{ in.})$
- **Provided Area ($A$):** $81.0 \text{ ft}^2$
- **Actual Soil Bearing Pressure ($q_{\text{act}}$):**

$$q_{\text{act}} = \frac{121.22 \text{ kips}}{81.0 \text{ ft}^2} = 1.497 \text{ ksf} \leq 1.50 \text{ ksf} \quad \text{(Safe)}$$

## PROMPT

Select the size of the area and the thickness of a spread footing to support an interior steel column.
The axial load on the column is 121.22 kips.
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Use IBC basic load combinations.

## Find Foundation using Google Gemini

### 2. Footing Thickness ($h$) Design

The thickness is determined using ACI 318 / IBC strength design principles based on shear capacity (two-way punching shear and one-way beam shear) and minimum cover requirements.

Assume concrete compressive strength $f'_c = 3,000$ psi and an ultimate factored load $P_u \approx 1.4 \times P = 169.7$ kips.

### A. Two-Way (Punching) Shear Check

The critical section for punching shear is at a distance $d/2$ from the perimeter of the $20$ in. $\times$ $20$ in. pedestal.

- **Critical Perimeter ($b_o$):** $4 \times (20 + d)$ in.
- **Concrete Punching Capacity ($\phi V_c$):**

$$\phi V_c = \phi \cdot 4\sqrt{f'_c}\, b_o d = 0.75 \times 4\sqrt{3000} \times 4(20 + d)d = 657.3(20 + d)d \text{ lbs}$$

Equating $\phi V_c$ to the ultimate factored load $P_u = 169,700$ lbs:

$$(20 + d)d = \frac{169,700}{657.3} \approx 258.2$$

$$d^2 + 20d - 258.2 = 0 \implies d \approx 8.9 \text{ in.}$$

## PROMPT

Select the size of the area and the thickness of a spread footing to support an interior steel column.
The axial load on the column is 121.22 kips.
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The column is supported by a 20-inch x 20-inch concrete pedestal resting on the footing.
Assume the concrete bearing capacity is 3,000 psf.
The allowable bearing pressure of the soil is 1,500 psf.
Use IBC basic load combinations.

**Note: the frost depth should be 60” for a Library in Houghton, Michigan (42” for Pinconning & Ypsilanti)**

**Foundation: 9’ x 9’ x 15” thickness**

B. Total Thickness Determination

- **Required Effective Depth ($d$):** $\approx 9.0$ in.
- **Clear Concrete Cover (Cast against earth):** $3.0$ in. (per ACI 318)
- **Assumed Bar Diameter (#6 bar):** $0.75$ in.

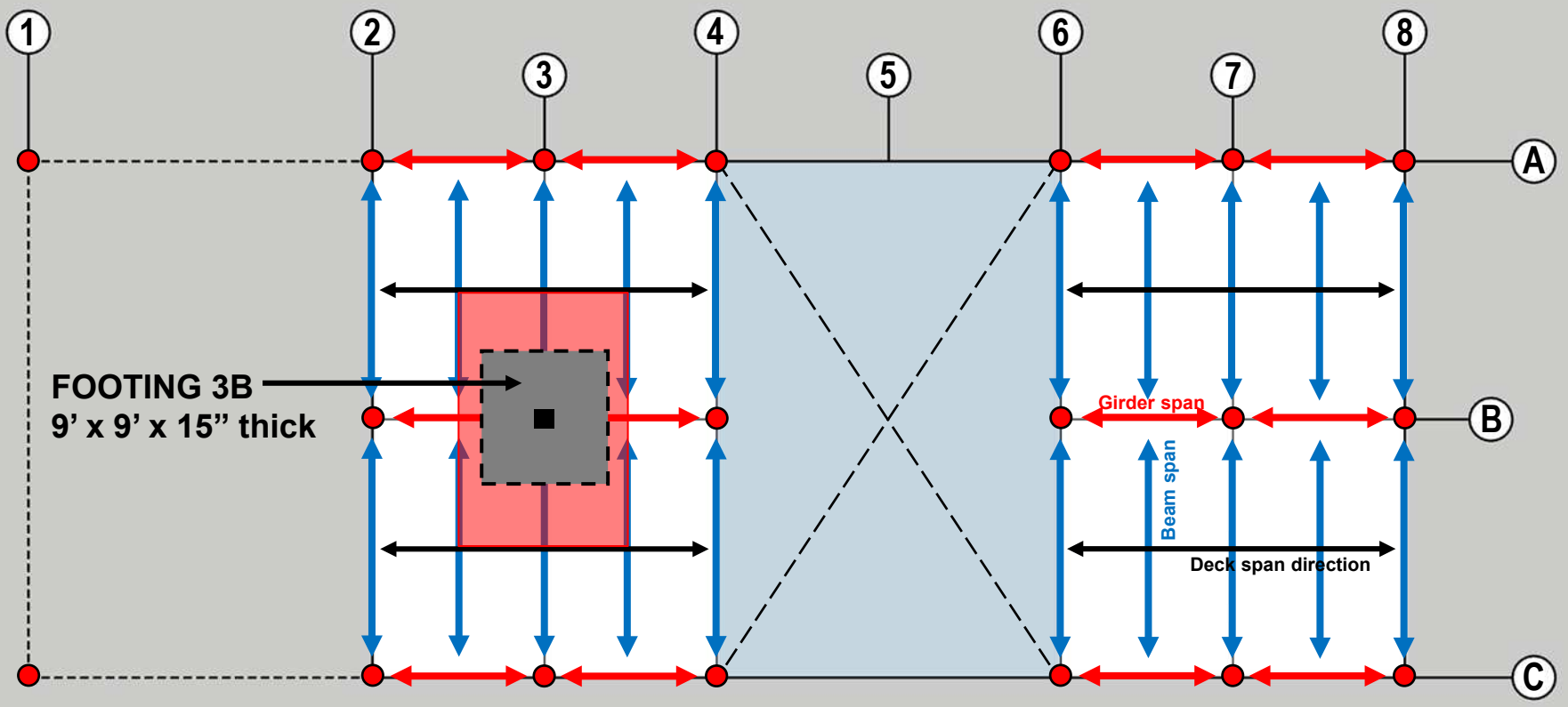
$$\text{Required } h = d + \text{Cover} + d_{\text{bar}} = 9.0 + 3.0 + 0.75 = 12.75 \text{ in.}$$

Accounting for the ACI minimum thickness requirement ($h \geq 12$ in. above bottom reinforcement) and standard nominal sizing increment:

- **Selected Footing Thickness ($h$):** $15$ in. ($1.25$ ft)
- **Provided Effective Depth ($d$):** $15 - 3.75 = 11.25$ in.

3. Summary of Footing Selection

| Parameter | Selected Design Value |
|---|---|
| **Footing Plan Dimensions** | $9.0$ ft $\times$ $9.0$ ft ($108$ in. $\times$ $108$ in.) |
| **Footing Total Area** | $81.0$ sq ft |
| **Footing Thickness ($h$)** | $15$ in. |
| **Depth Below Grade ($D_f$)** | $42$ in. ($3.5$ ft) |
| **Pedestal Size** | $20$ in. $\times$ $20$ in. |




**<u>COLUMN 3B</u> (Floor-Load only)**
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**121.22 kips = Column 3B Axial Load**
12.25’ = Column 3B Height
ASTM A992 Steel (50 ksi Yield Strength)
**W8 x 28: Column 3B**

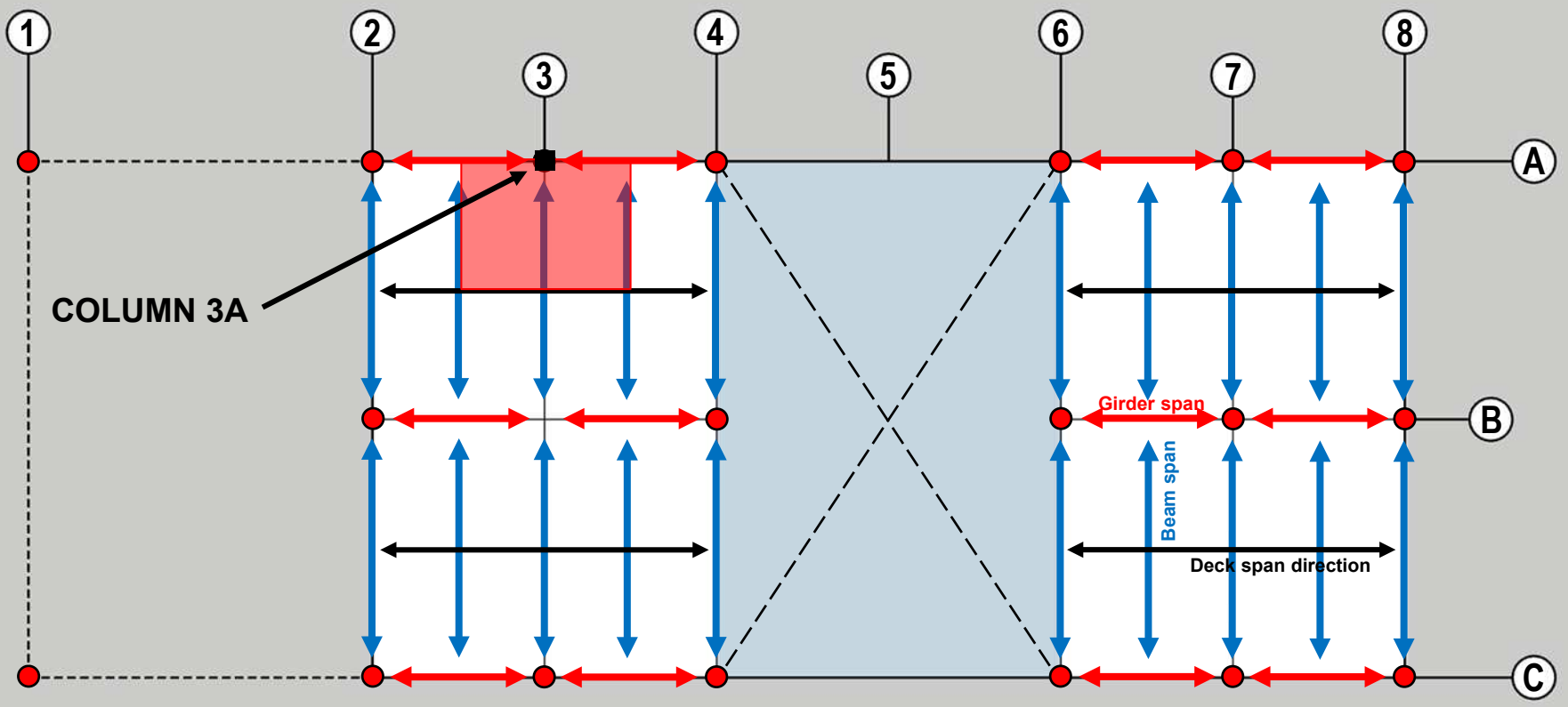
**<u>FOUNDATION 3B</u> Spread Footing with 20” x 20” Column Pedestal**
**Footing Depth Below Grade: 60”**
**Footing Area = 81 sf** = 9’ x 9’
**Footing Thickness = 1’-3”** = 1.25” = 15”
**Sized using Google Gemini with Prompt:**
*Select the size of the area and the thickness of a spread footing to support an interior steel column. The axial load on the column is 121.22 kips. The column is a W8 x 28 section.*
*The depth of the footing must be 42 inches below grade. Assume the concrete bearing capacity is 3,000 psf.*
*The column is supported by a 20-inch x 20-inch concrete pedestal resting on the footing.*
*The allowable bearing pressure of the soil is 1,500 psf. Use IBC basic load combinations.*

**9’ x 9’ x 15” Thick: Spread Footing @ 3B**

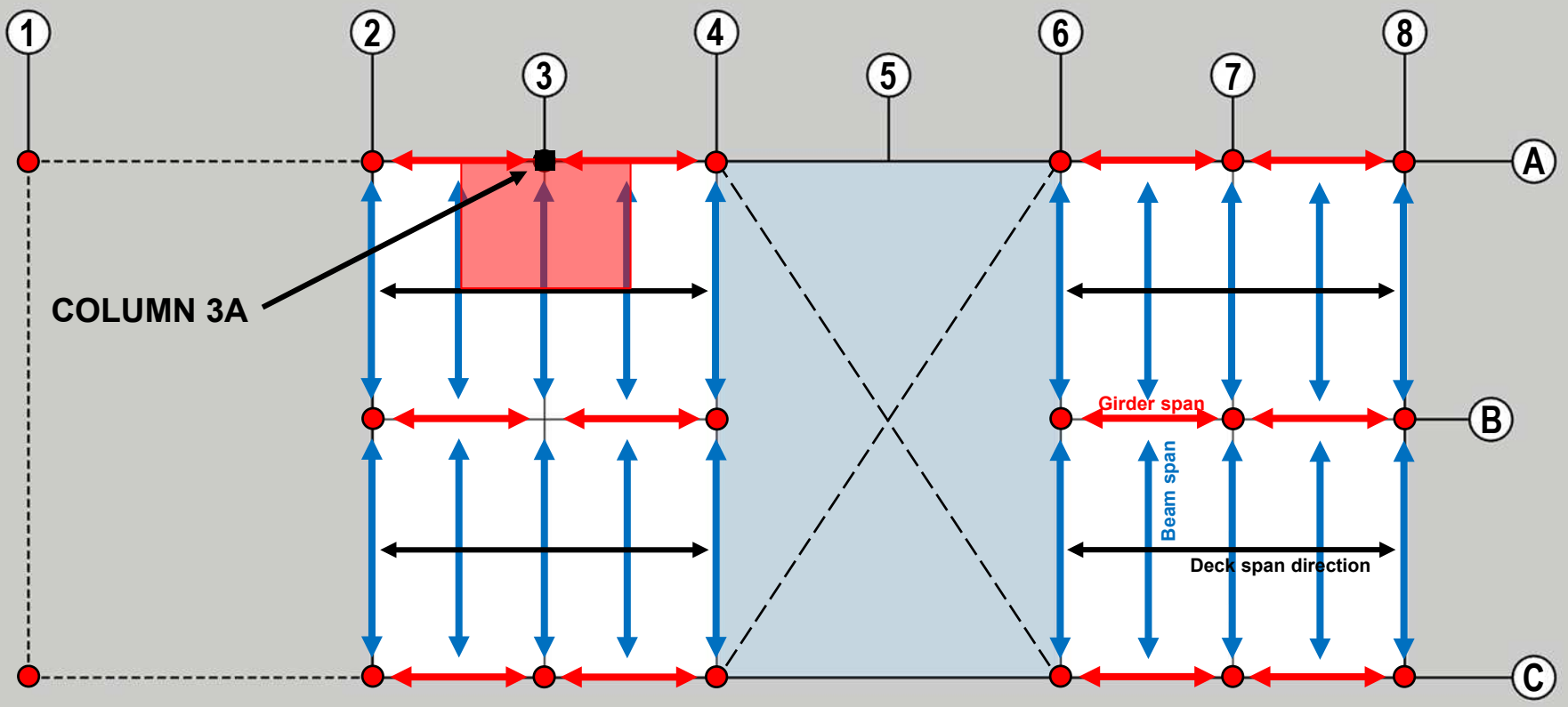
# COLUMN 3B and FOUNDATION 3B

*…find the column size of Column 3A*

**COLUMN 3A**




**COLUMN 3A*…find the axial load on column 3A…(floor load only)***




**<u>COLUMN 3A</u> (Floor-Load only)**
**50% of Load on Column 3B = 60.61 kips** = 121.22 kips (.5)

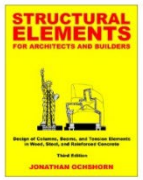
**60.61 kips = Column 3A Axial Load**
12.25' = Column 3B Height
ASTM A992 Steel (50 ksi Yield Strength)

# COLUMN 3A*…find the axial load on column 3A…(floor load only)*

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## Steel wide-flange column design calculator

Jonathan Ochshorn

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**Steel Type: A992**
This type has ultimate yield strength of 50 ksi

**Height: The unbraced height of the column**
Also know as the effective length

**K: The effective length factor**
This factor influences the unbraced height and the connection type at the top and bottom of the column.
**K = 1** for a steel system with lateral-load resistant braced frames and roof/floor diaphragms.

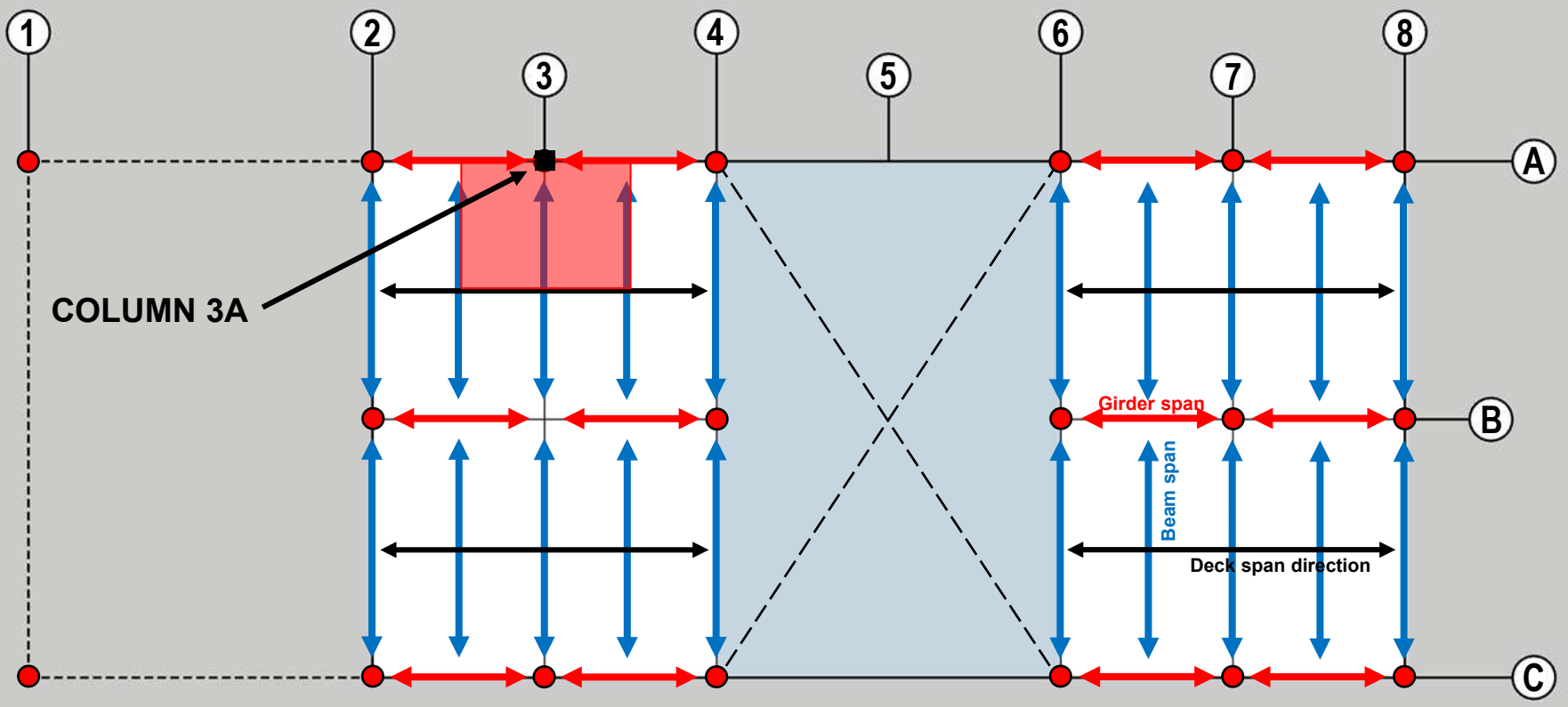
**Axial Load: The load applied to the column**

| | | |
|---|---|---|
| Steel type | A992 | |
| Height | 12.25 | feet |
| K | 1 | |
| Axial load | 60.61 | kips |
| Limit KL/r to 200 | | |

**Five lightest choices:**

| | |
|---|---|
| 1 | ***W6x15*** |
| 2 | W5x19 |
| 3 | W6x20 |
| 4 | W8x21 |
| 5 | W10x22 |

**Column Size = W6 x 15**

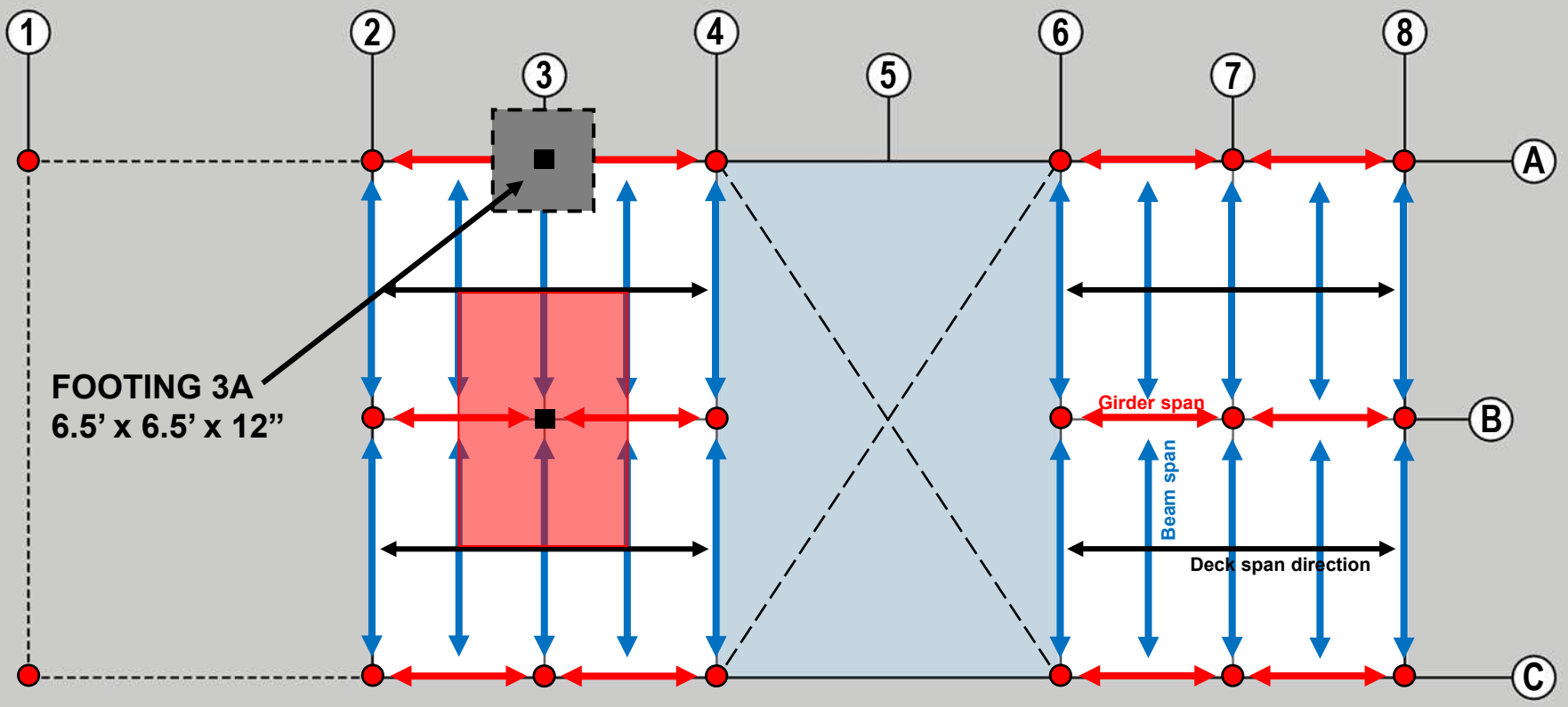



**<u>COLUMN 3A</u> (Floor-Load only)**
**50% of Load on Column 3B = 60.61 kips** = 121.22 kips (.5)

**60.61 kips = Column 3A Axial Load**
12.25' = Column 3B Height
ASTM A992 Steel (50 ksi Yield Strength)

**W6 x 15: Column 3A**

## COLUMN 3A and FOUNDATION 3A*…what size is the foundation?*




**<u>COLUMN 3A</u> (Floor-Load only)**
**50% of Load on Column 3B = 60.61 kips** = 121.22 kips (.5)

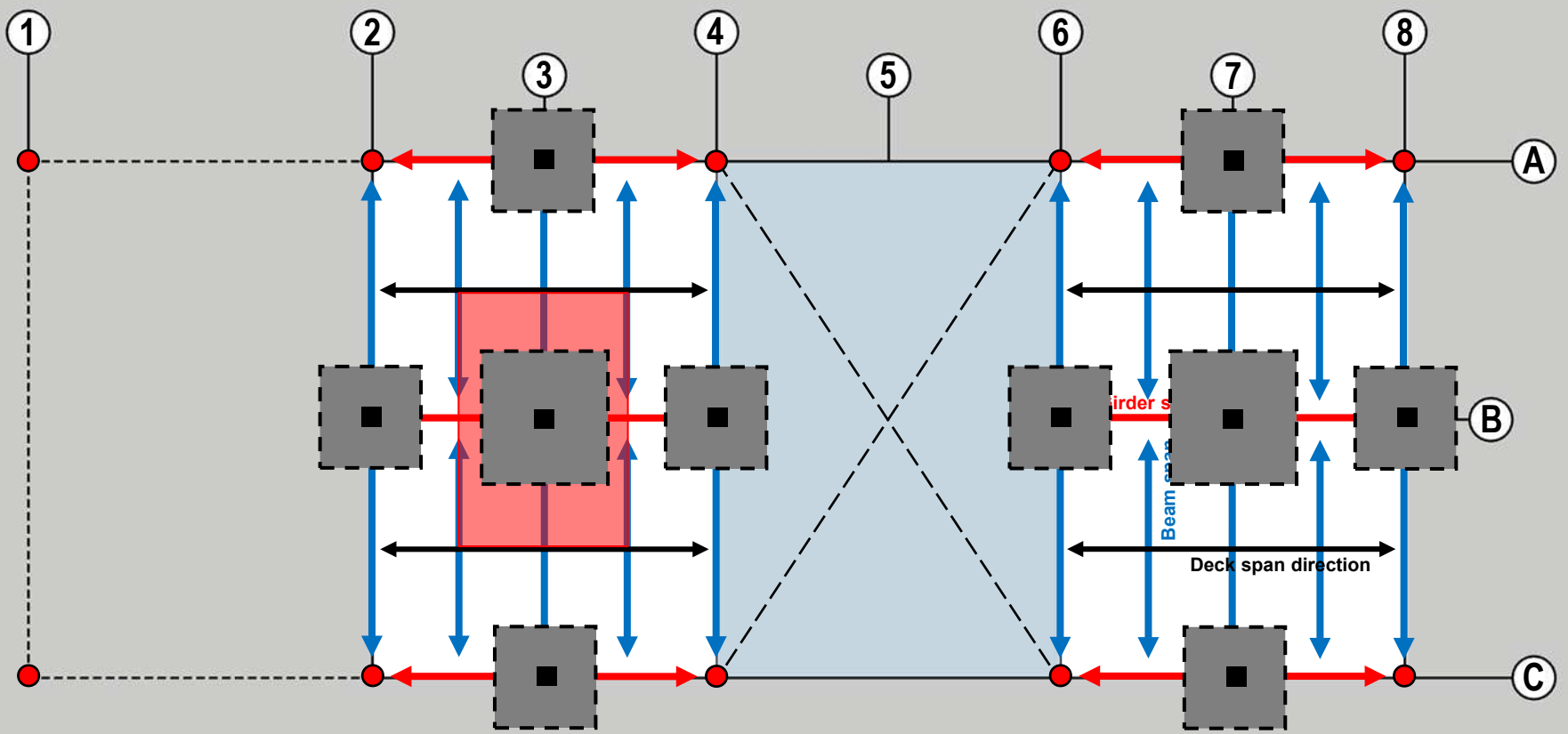
**60.61 kips = Column 3A Axial Load**
12.25’ = Column 3B Height
ASTM A992 Steel (50 ksi Yield Strength)

**W6 x 15: Column 3A**

**<u>FOUNDATION 3A</u> Spread Footing with 20” x 20” Column Pedestal**
**Footing Depth Below Grade: 60”**
**Footing Area = 42.25 sf** = 6.5’ x 6.5’
**Footing Thickness = 1’-0”** = 12”
**Sized using Google Gemini with Prompt:**
*Select the size of the area and the thickness of a spread footing to support an interior steel column. The axial load on the column is 60.61 kips. The column is a W6 x 15 section.*
*The depth of the footing must be 60 inches below grade. Assume the concrete bearing capacity is 3,000 psf.*
*The column is supported by a 20-inch x 20-inch concrete pedestal resting on the footing.*
*The allowable bearing pressure of the soil is 1,500 psf. Use IBC basic load combinations.*

**6.5’ x 6.5’ x 12” Thick: Spread Footing @ 3A**

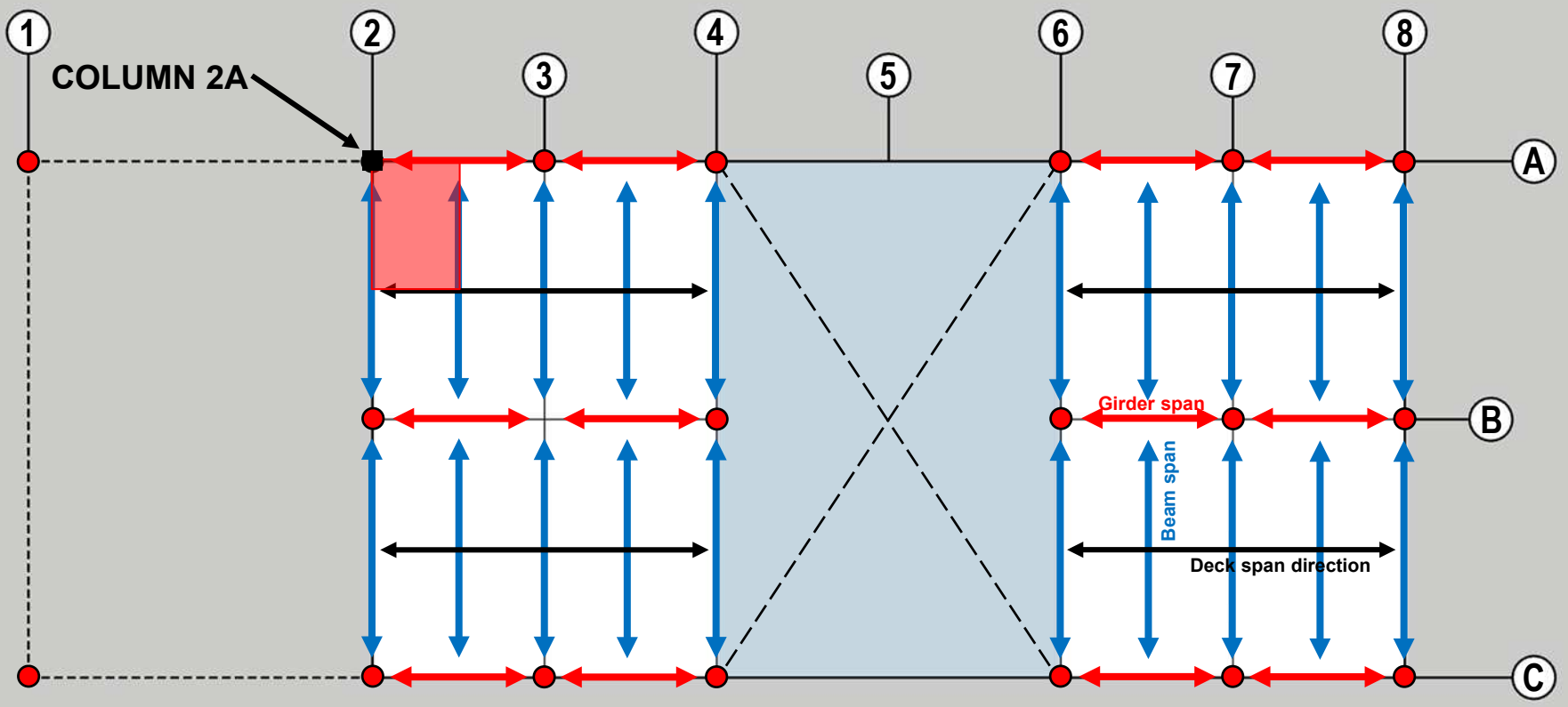
# COLUMN 3B and FOUNDATION 3B



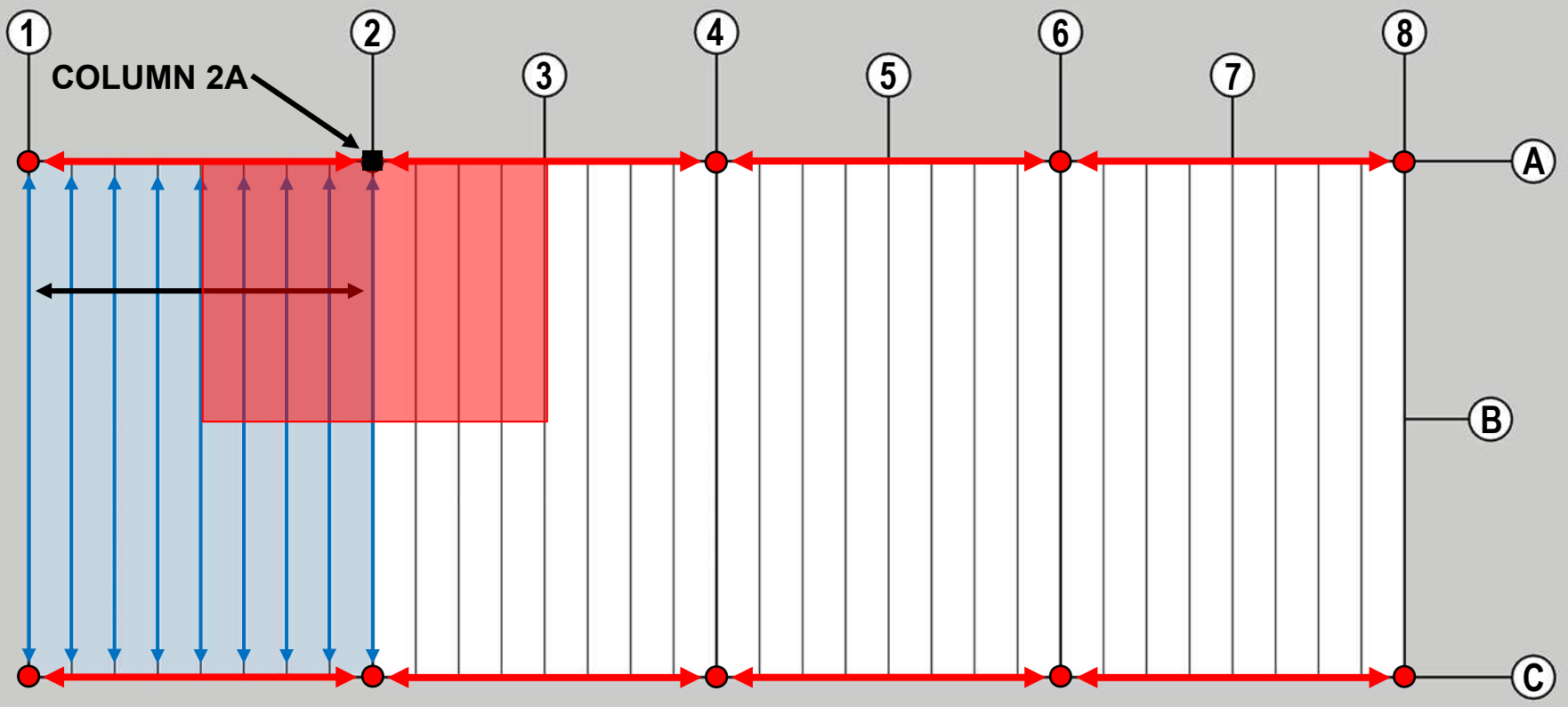

# FOUNDATION FOR COLUMNS SUPPORTING FLOOR LOAD ONLY

*…find the column and foundation size of Column 2A*

**COLUMN 2A**




**COLUMN 2A***…find column and foundation size?*




**<u>COLUMN 2A</u> (Roof-Load)**
**Roof Deck Tributary Area = 1.200 sf** = 40’ x 30’
**Roof Load = 117 psf** = (100 psf + 15 psf + 2psf)
**Roof Deck Tributary Load = 140.4 kips** = (117 psf) (1,200 sf)
**Open-Web Joists Self-weight = 0.897 kips** = 29.9 plf (30’ length of 40LH13)
**Joist Girder Self-weight = 3.12 kips** = 78 plf (40’ length of 44G7N24)
**Roof Tributary Load on Column 2A = 144.42 kips**
(140.4 kips + .897 kips + 3.12 kips)

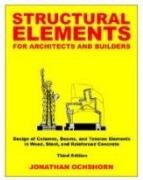
**144.42 kips = Column 2A Axial Load**
16’ = Column 2A Height
ASTM A992 Steel (50 ksi Yield Strength)

# ROOF LOAD ON COLUMN 2A

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## Steel wide-flange column design calculator

Jonathan Ochshorn

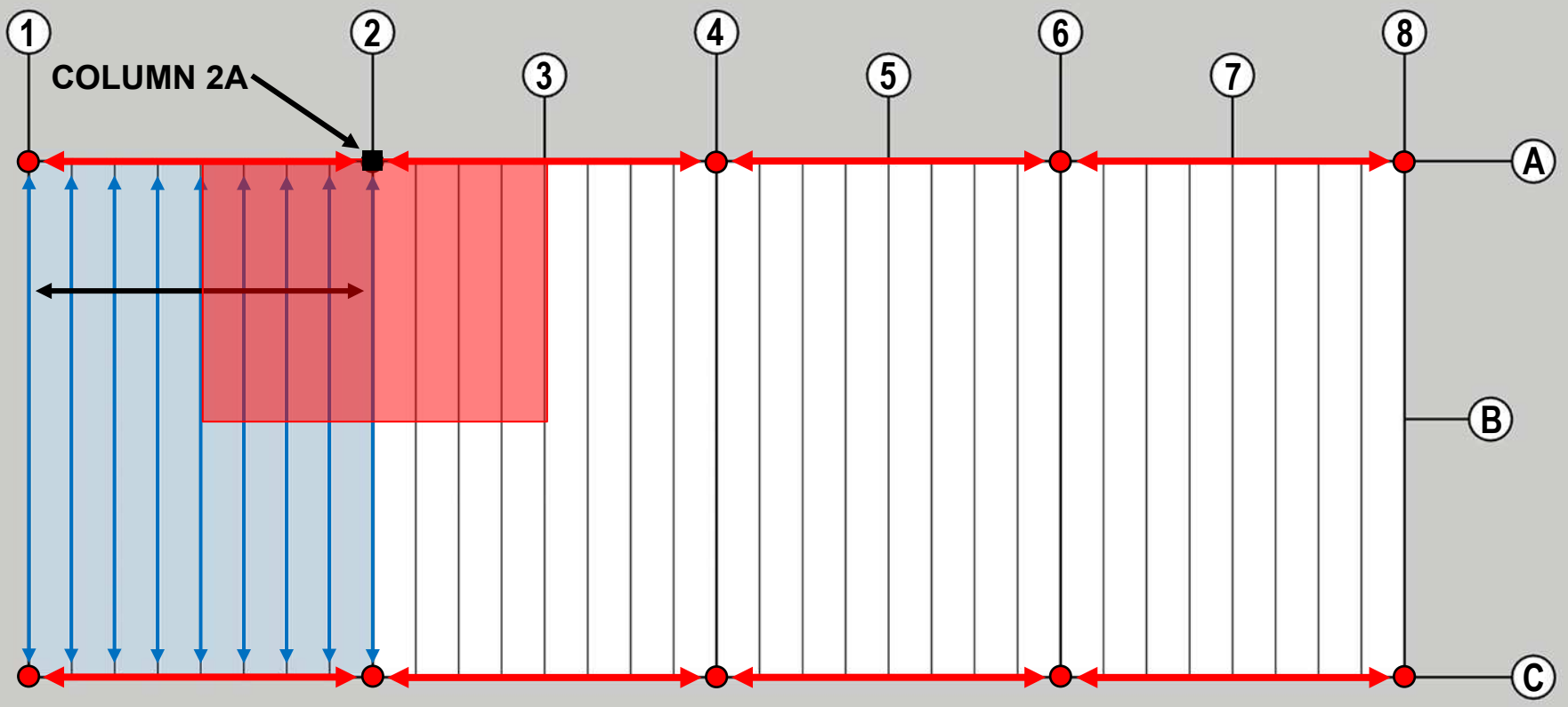
© 2016 Jonathan Ochshorn.

| | | |
|---|---|---|
| Steel type | A992 | |
| Height | 15 | feet |
| K | 1 | |
| Axial load | 144.42 | kips |
| Limit KL/r to 200 | | |

**Five lightest choices:**

| | |
|---|---|
| 1 | ***W8x31*** |
| 2 | W10x33 |
| 3 | W8x35 |
| 4 | W10x39 |
| 5 | W12x40 |

**Column Size = W8 x 31**

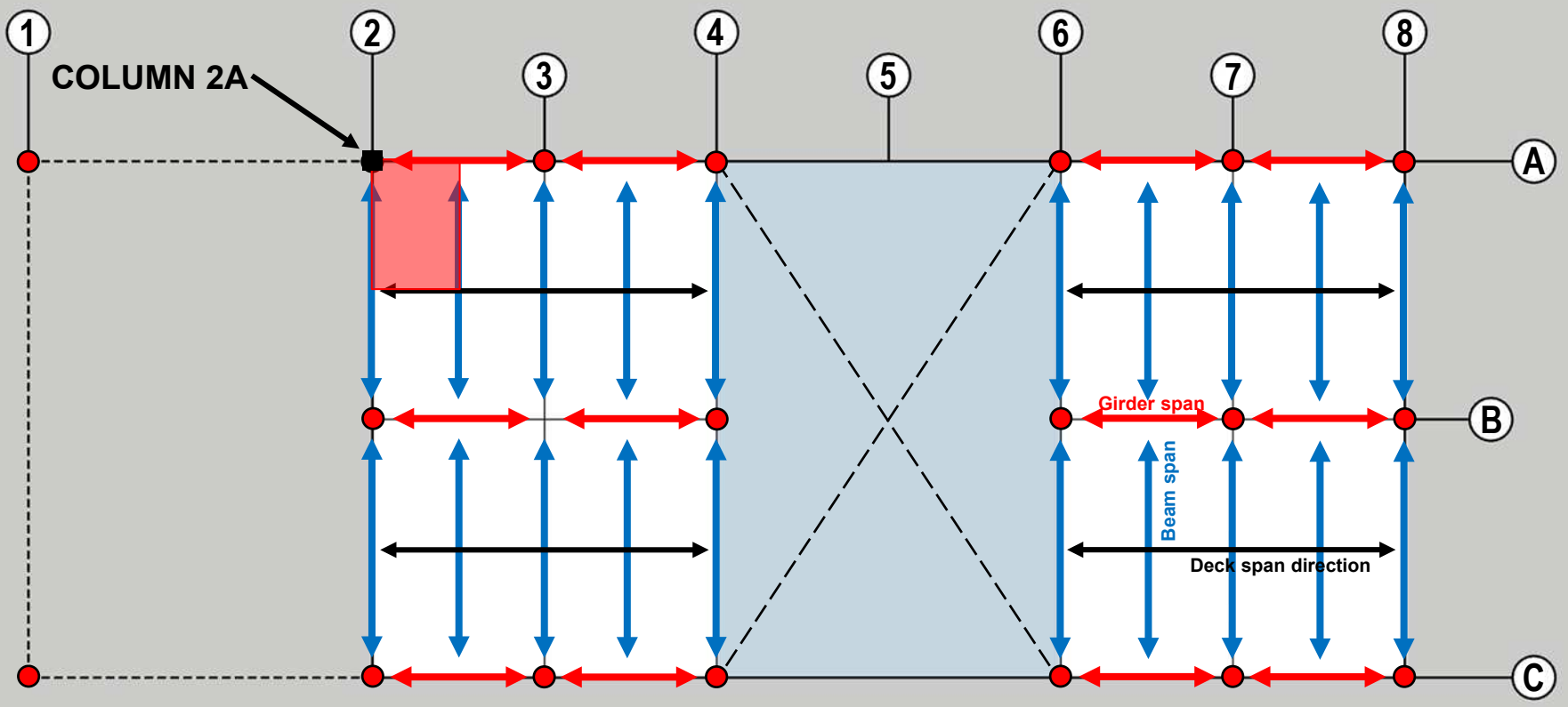



**<u>COLUMN 2A</u> (Roof-Load)**

**144.42 kips = Column 2A Axial Load**
16’ = Column 2A Height
ASTM A992 Steel (50 ksi Yield Strength)

**W8 X 31: COLUMN 2A for roof load**

# COLUMN 2A

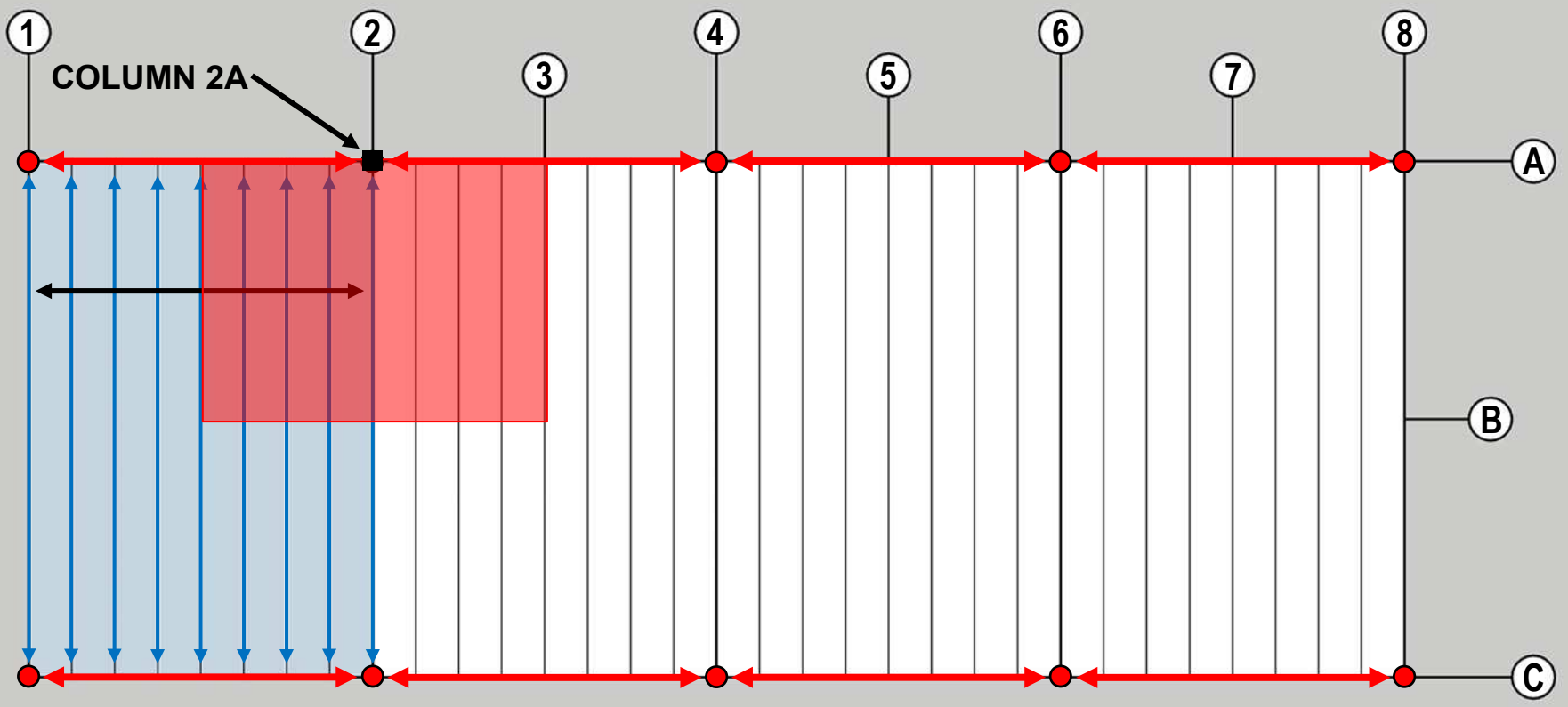



**<u>COLUMN 2A</u> (Floor-Load only)**
**25% of Load on Column 3B = 30.3 kips** = 121.22 kips (.25)

**30.3 kips = Column 2A Axial Load**
12.25' = Column 3B Height
ASTM A992 Steel (50 ksi Yield Strength)

**W6 x 15: Column 2A for floor load ????**

## COLUMN 2A*…column load smaller at floor?*




**COLUMN 2A (Roof-Load)**

**144.42 kips = Column 2A Axial Load**
16' = Column 2A Height
ASTM A992 Steel (50 ksi Yield Strength)

**W8 X 31: COLUMN 2A for roof load only**

**COLUMN 2A (Floor-Load only)**
**25% of Load on Column 3B = 30.3 kips** = 121.22 kips (.25)

**30.3 kips = Column 2A Axial Load Floor only**
12.25' = Column 2A Height
ASTM A992 Steel (50 ksi Yield Strength)

**COLUMN 2A (Roof-Load + Floor-Load)**
**25% of Load on Column 3B = 30.3 kips** = 121.22 kips (.25)

**30.3 kips = Column 2A Axial Floor Load**
**144.42 kips = Column 2A Axial Roof Load**

**174.72 kips = Column 2A Axial Load**
12.25' = Column 2A Height unbraced
ASTM A992 Steel (50 ksi Yield Strength)

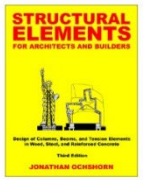
**W8 x 31: Column 2A for total load & height**

# COLUMN 2A

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## Steel wide-flange column design calculator

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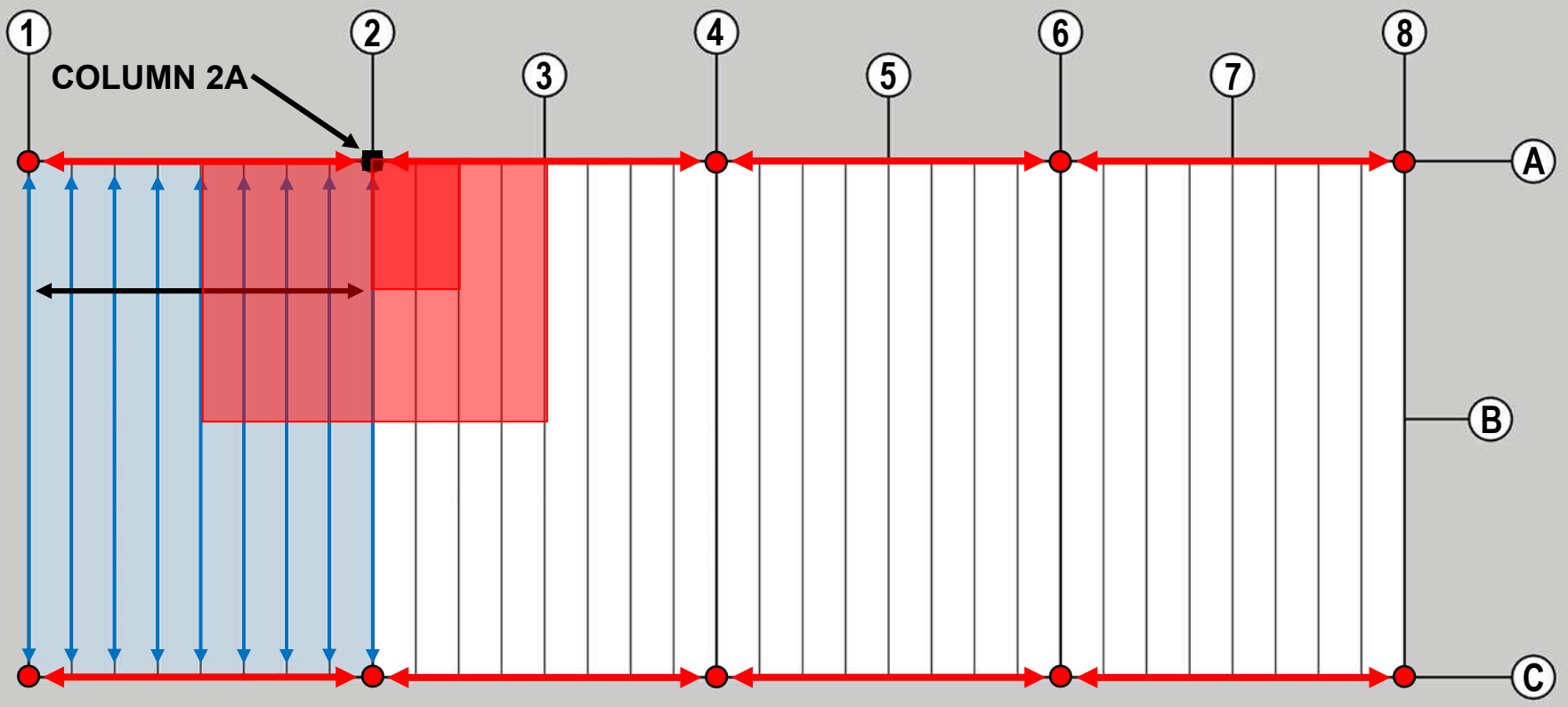
© 2016 Jonathan Ochshorn.

| | | |
|---|---|---|
| Steel type | A992 | |
| Height | 12.25 | feet |
| K | 1 | |
| Axial load | 174.72 | kips |
| Limit KL/r to 200 | | |

**Five lightest choices:**

| | |
|---|---|
| 1 | ***W8x31*** |
| 2 | W10x33 |
| 3 | W8x35 |
| 4 | W10x39 |
| 5 | W12x40 |

**Column Size = W8 x 31**



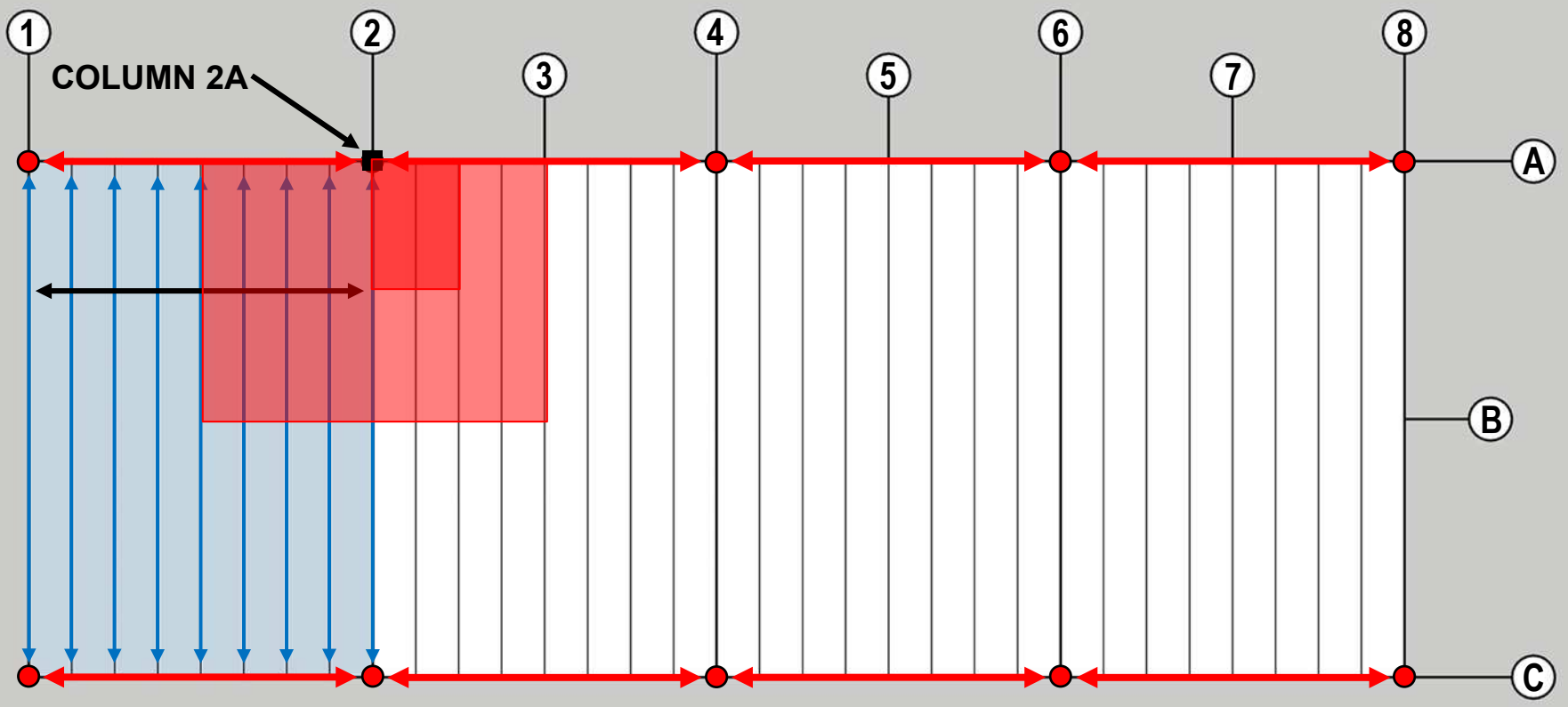

**<u>COLUMN 2A</u> (Roof-Load + Floor-Load)**
**25% of Load on Column 3B = 30.3 kips** = 121.22 kips (.25)

**30.3 kips = Column 2A Axial Floor Load**
**144.42 kips = Column 2A Axial Roof Load**

**174.72 kips = Column 2A Axial Load**
12.25' = Column 2A Height unbraced
ASTM A992 Steel (50 ksi Yield Strength)

**W8 x 31: Column 2A for total load & height**

# COLUMN 2A & FOUNDATION 2A




**<u>COLUMN 2A</u> (Roof-Load + Floor-Load)**
**25% of Load on Column 3B = 30.3 kips** = 121.22 kips (.25)

**30.3 kips = Column 2A Axial Floor Load**
**144.42 kips = Column 2A Axial Roof Load**

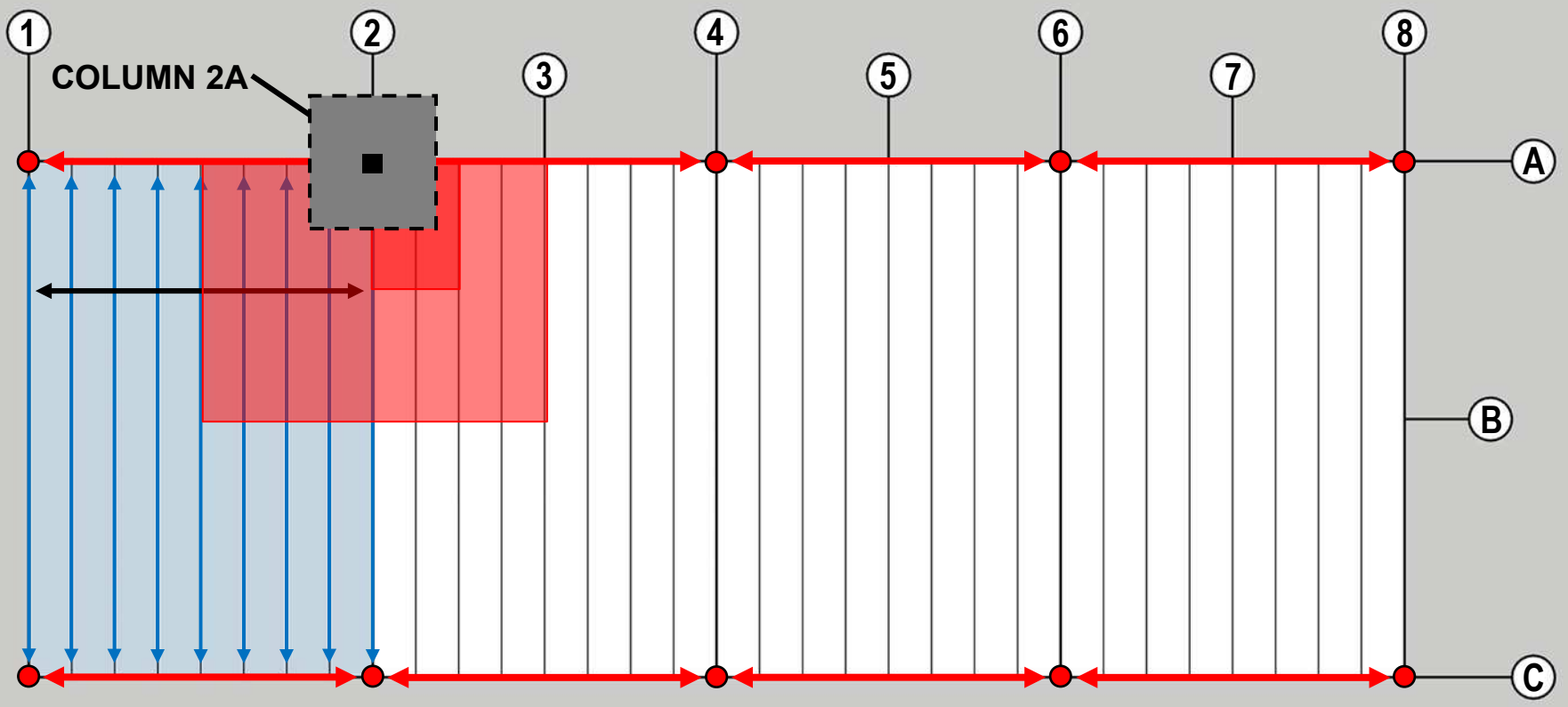
**174.72 kips = Column 2A Axial Load**
12.25' = Column 2A Height unbraced
ASTM A992 Steel (50 ksi Yield Strength)

**W8 x 31: Column 2A for total load & height**

**<u>FOUNDATION 2A</u> Spread Footing with 20" x 20" Column Pedestal**
**Footing Depth Below Grade: 60"**
**Footing Area = 42.25 sf** = 6.5' x 6.5'
**Footing Thickness = 1'-0"** = 12"
**Sized using Google Gemini with Prompt:**
*Select the size of the area and the thickness of a spread footing to support an interior steel column. The axial load on the column is 174.72 kips. The column is a W8 x 31 section.*
*The depth of the footing must be 60 inches below grade. Assume the concrete bearing capacity is 3,000 psf.*
*The column is supported by a 20-inch x 20-inch concrete pedestal resting on the footing.*
*The allowable bearing pressure of the soil is 1,500 psf. Use IBC basic load combinations.*

**11' x 11' x 18" Thick: Spread Footing @ 2A**

# COLUMN 2A & FOUNDATION 2A




**<u>COLUMN 2A</u> (Roof-Load + Floor-Load)**
**25% of Load on Column 3B = 30.3 kips** = 121.22 kips (.25)

**30.3 kips = Column 2A Axial Floor Load**
**144.42 kips = Column 2A Axial Roof Load**

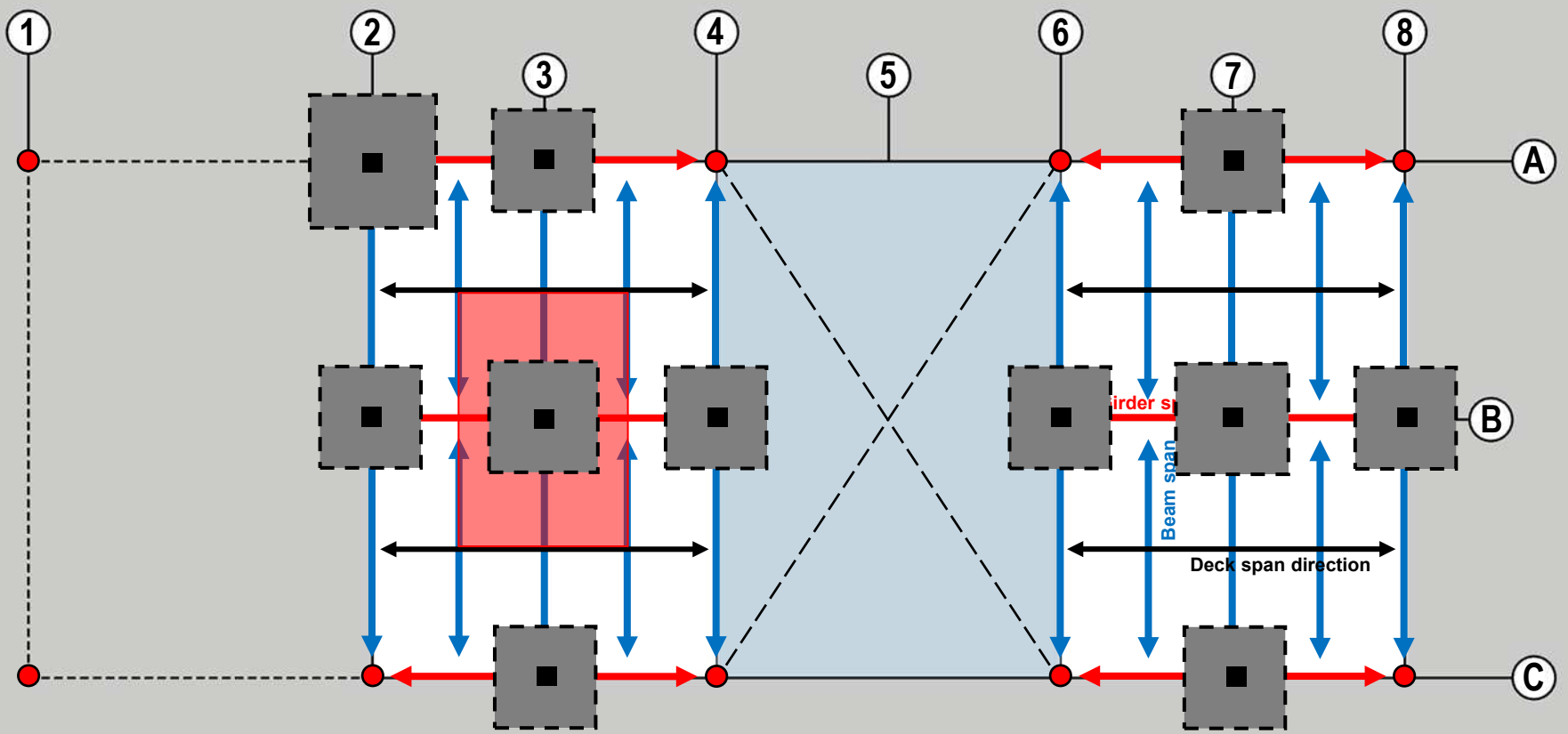
**174.72 kips = Column 2A Axial Load**
12.25' = Column 2A Height unbraced
ASTM A992 Steel (50 ksi Yield Strength)

**W8 x 31: Column 2A for total load & height**

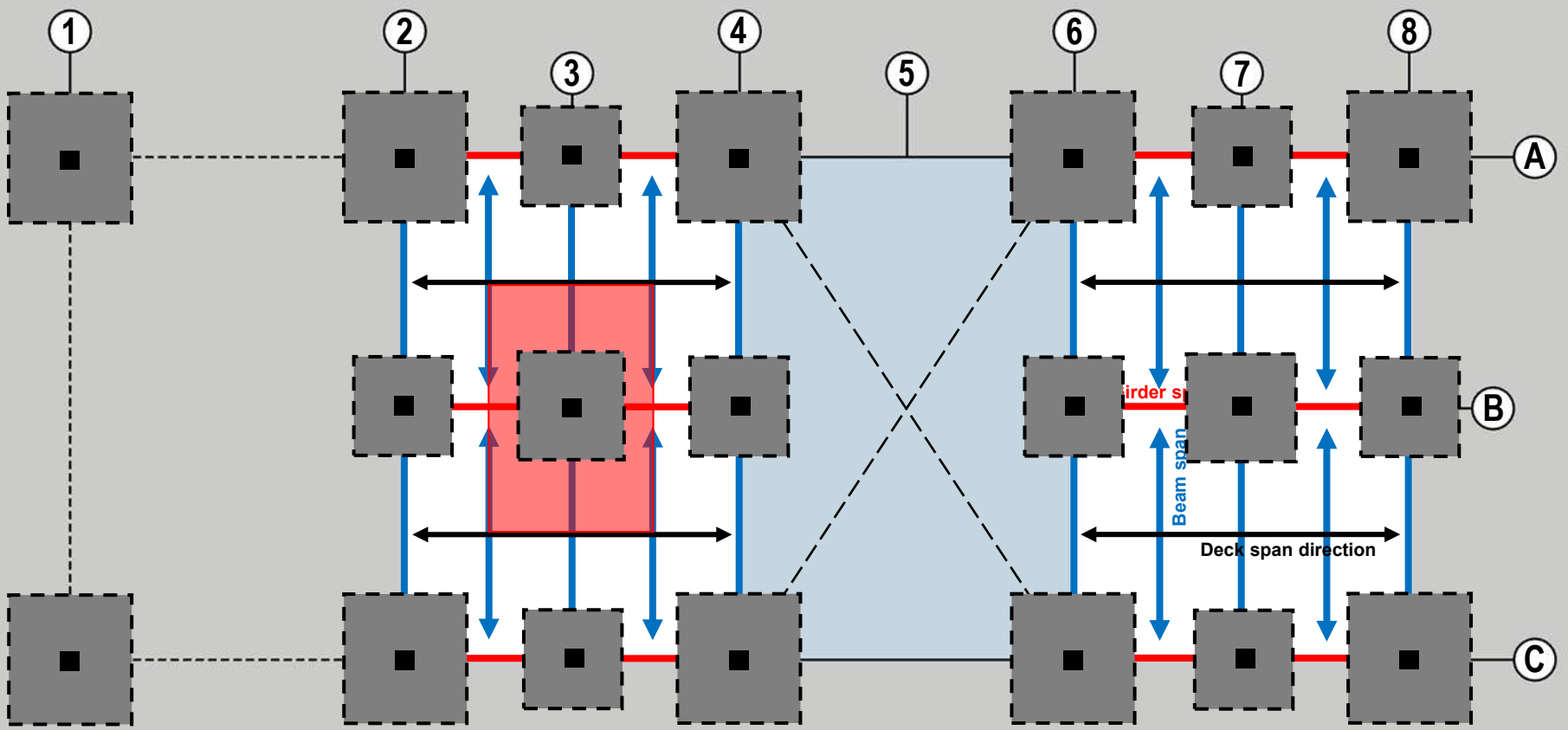
**<u>FOUNDATION 2A</u> Spread Footing with 20" x 20" Column Pedestal**
**Footing Depth Below Grade: 60"**
**Footing Area = 42.25 sf** = 6.5' x 6.5'
**Footing Thickness = 1'-0"** = 12"
**Sized using Google Gemini with Prompt:**
*Select the size of the area and the thickness of a spread footing to support an interior steel column. The axial load on the column is 174.72 kips. The column is a W8 x 31 section.*
*The depth of the footing must be 60 inches below grade. Assume the concrete bearing capacity is 3,000 psf.*
*The column is supported by a 20-inch x 20-inch concrete pedestal resting on the footing.*
*The allowable bearing pressure of the soil is 1,500 psf. Use IBC basic load combinations.*

**11' x 11' x 18" Thick: Spread Footing @ 2A**

# COLUMN 2A & FOUNDATION 2A




# FOUNDATION FOR COLUMNS SUPPORTING FLOOR LOAD & 2A ROOF LOAD



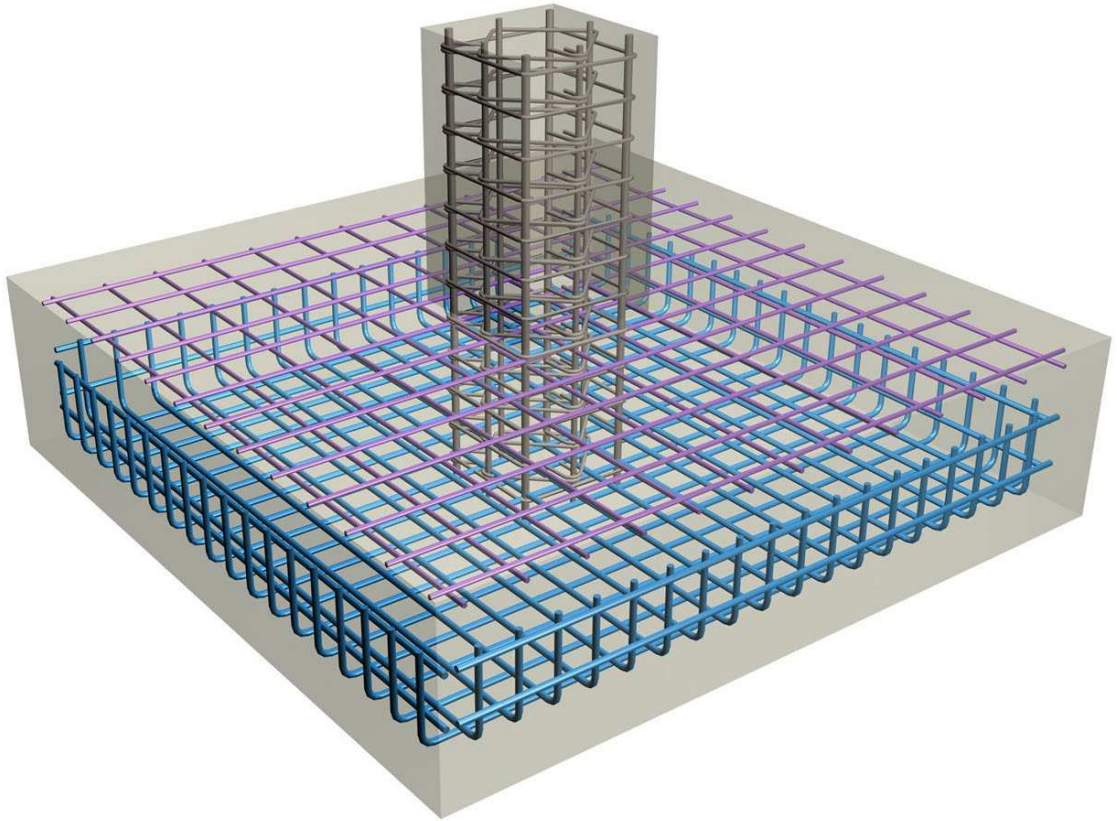

**FOUNDATION 1A, 1C, 8A, 8C**
**Footing Depth Below Grade: 60”**
**Footing Area = 42.25 sf** = 6.5’ x 6.5’
**Footing Thickness = 1’-0”** = 12”

**10’ x 10’ x 15” Thick: Spread Footing @ 1A, 1C, 8A, 8C**

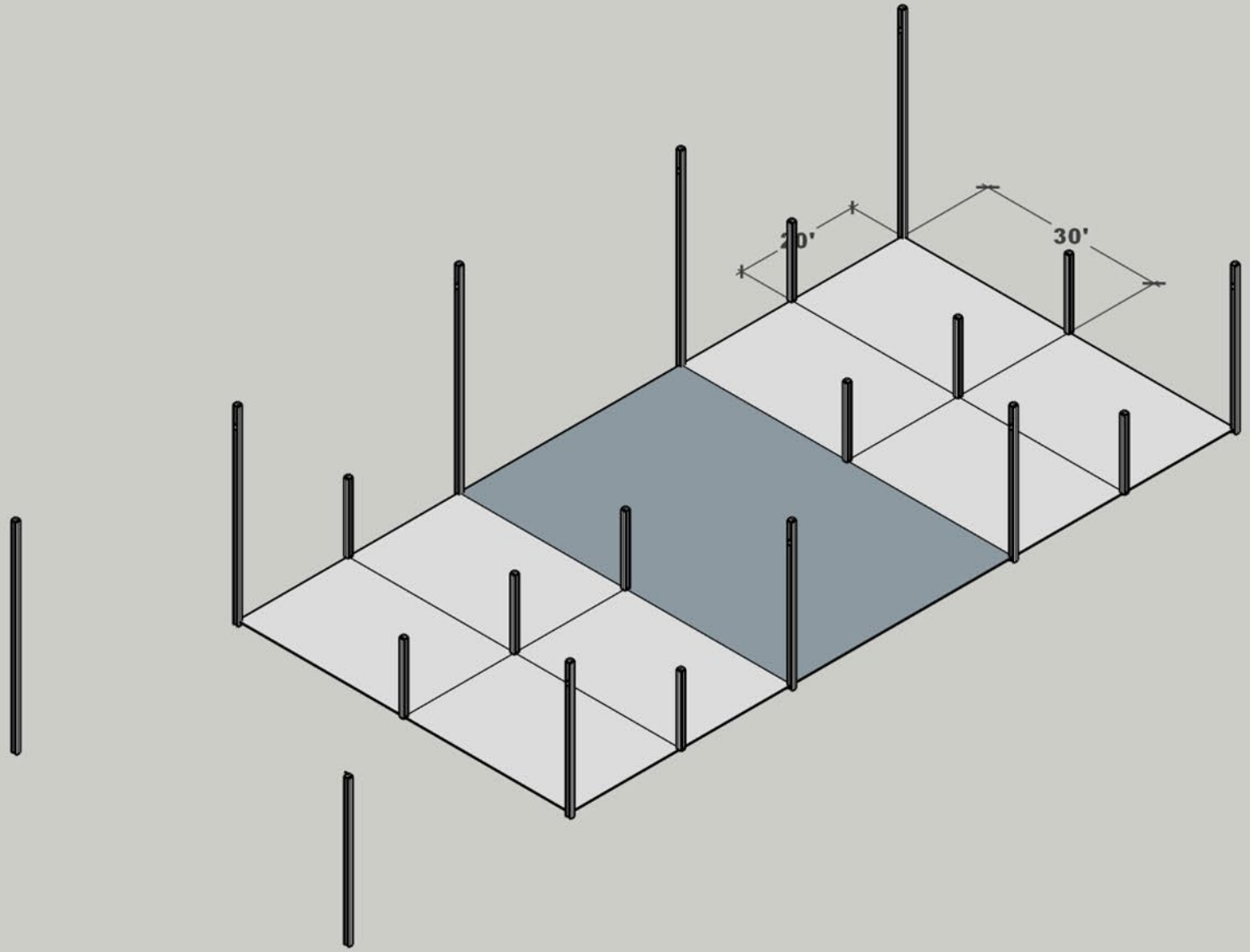
**Sized using Google Gemini with Prompt:**
*Select the size of the area and the thickness of a spread footing to support an interior steel column. The axial load on the column is 87.36 kips. The column is a W8 x 31 section. The depth of the footing must be 60 inches below grade. Assume the concrete bearing capacity is 3,000 psf. The column is supported by a 20-inch x 20-inch concrete pedestal resting on the footing. The allowable bearing pressure of the soil is 1,500 psf. Use IBC basic load combinations.*

# FOUNDATION PLAN FOR COLUMNS SUPPORTING FLOOR LOAD & ROOF LOAD



**SPREAD FOOTING WITH CONCRETE PEDESTAL FOR STEEL COLUMN**




*…add all foundations to your axon sequence*

# STEEL BUILDING PROJECT

*…questions on Columns or Foundation?*